

TTK4240 Industriell elektroteknikk

January 20, 2025

Marta Molinas

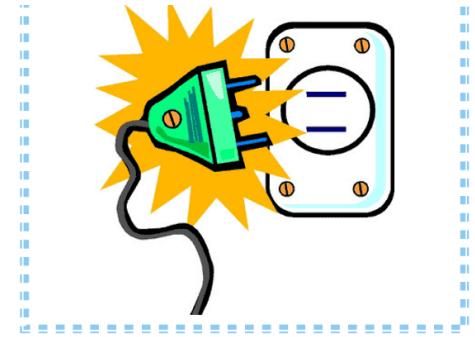
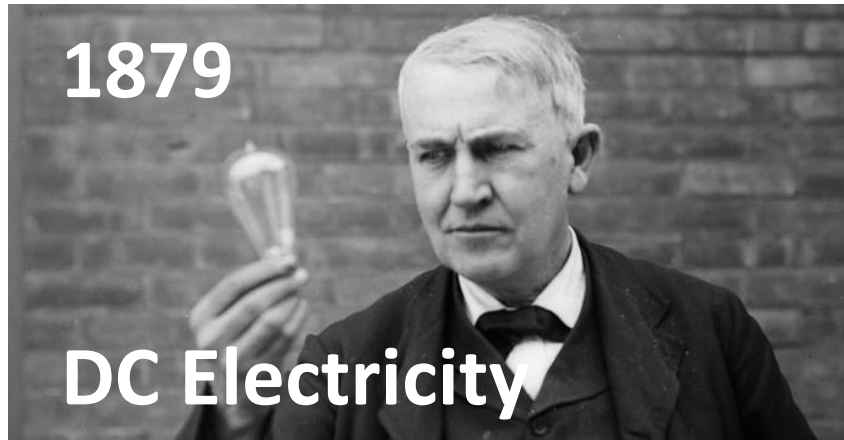
Institutt for teknisk kybernetikk

Lecture - Week 04 Part I

- Review of the fundamentals in circuit analysis with sinusoidal signals
 - The concept of Phasor
- Current, voltage and power calculations
 - Instantaneous power
 - Average (real) power
 - Reactive power
 - Complex power
 - Power factor
 - Maximum real power delivered to a load
 - Load impedance required to deliver maximum real power to the load

Nilsson's Book: 9.1 to 9.7, Chapter 10 Complete

The History of ELECTRICITY



Thomas Alva Edison first light bulb
National Museum of American
History



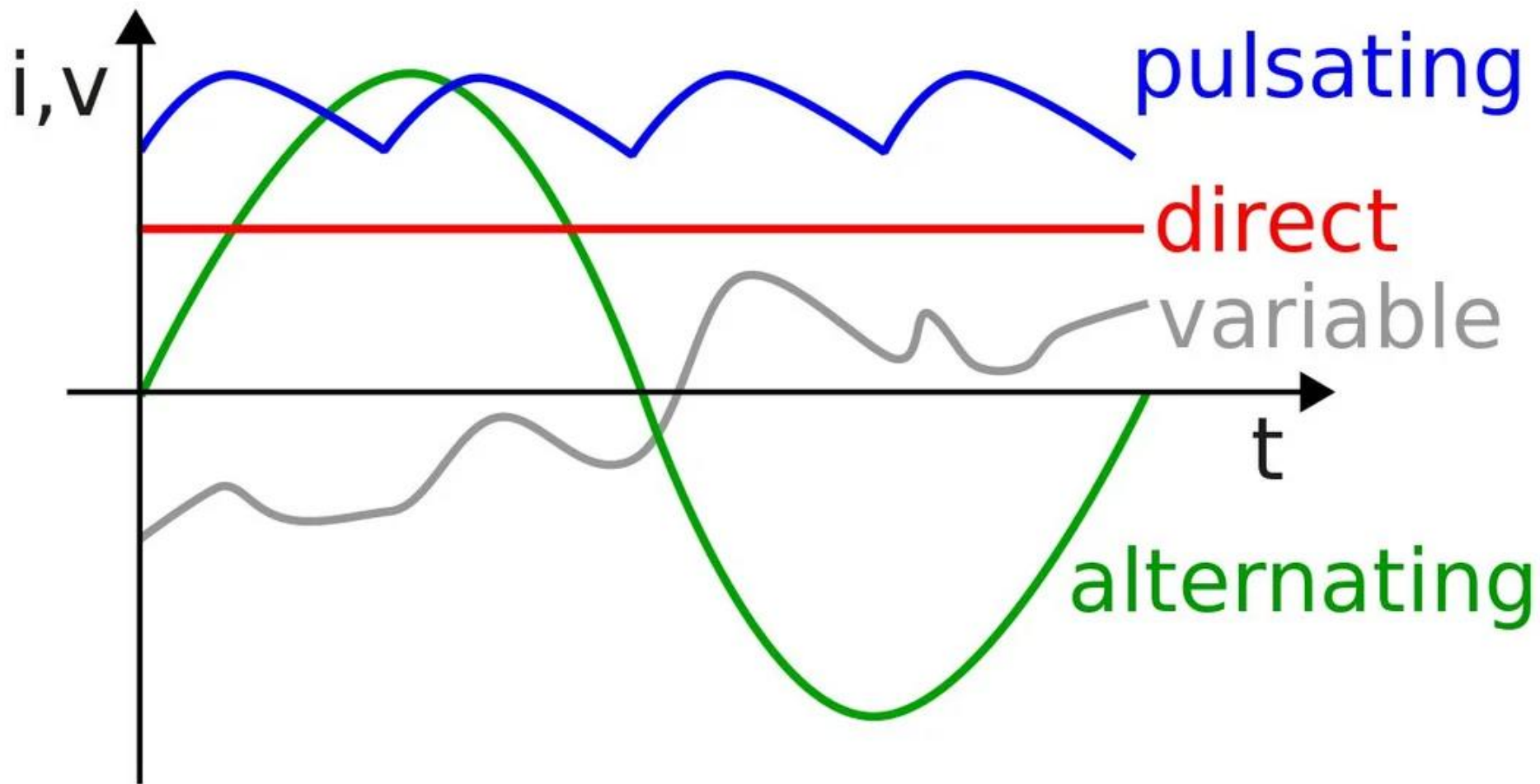
Tesla's AC Induction
Motor is one of the 10
greatest discoveries of
all time

1888
AC Electricity

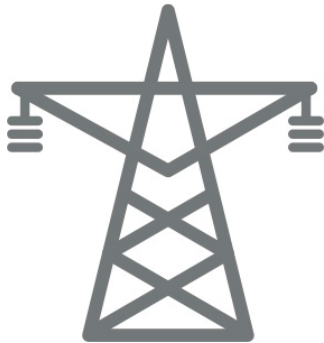
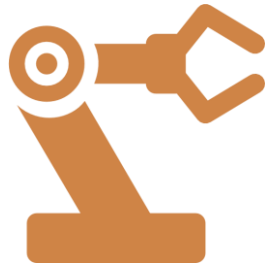
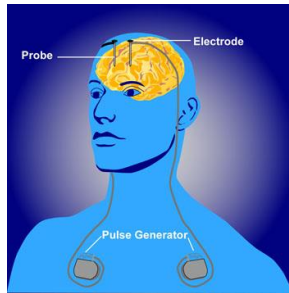
<https://www.youtube.com/watch?v=Cidg4Xfpjmc>

Smithsonian Museum
in Washington DC





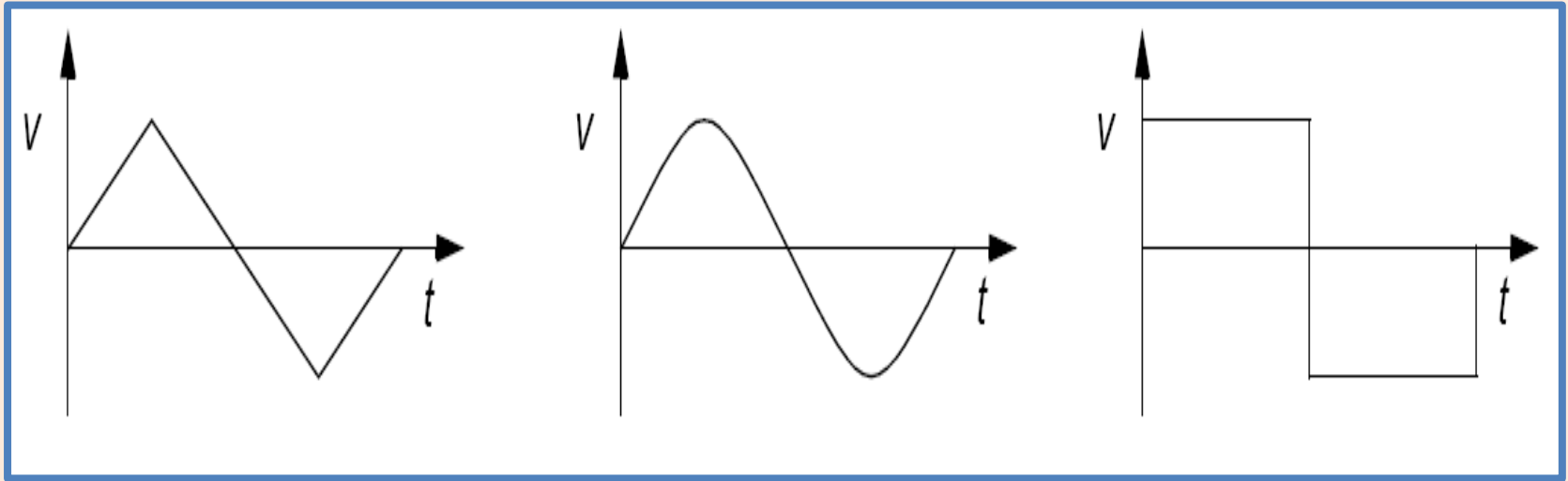
Electric Power uses



Review: Alternating Current (AC)

- Why AC?
- Voltage waveform?
- Choice of frequency (f)

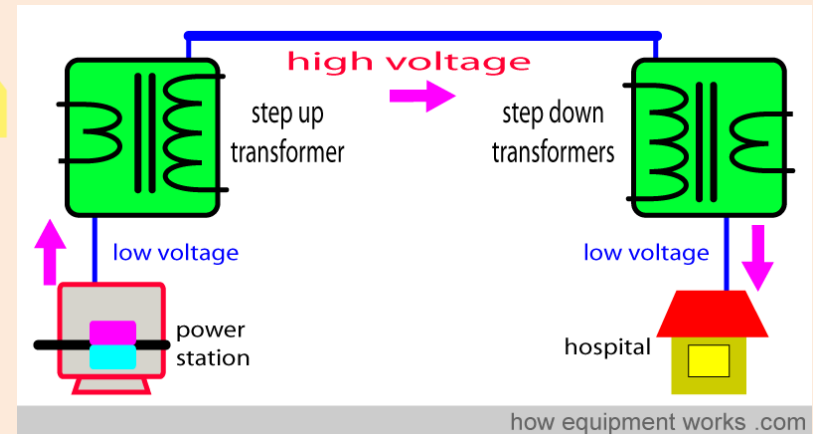
$$v(t) = A \sin(\omega t + \theta)$$



Why Alternating Current (AC)

- Virtually all power supply / distribution takes place with sinusoidal voltages
 - ❖ Possible to transform voltages (transformers: step up and step down)
 - ❖ Lower losses (with higher voltages)

$$P = V \times I \rightarrow I = \frac{P}{V}$$
$$\text{losses} = I^2 R$$



how equipment works .com

Why 50/60Hz?

- Low frequency in the range 10 or 20 Hz, will cause flicker

- A high frequency

- Increases hysteresis losses

$$P_h \propto f \cdot B^n$$

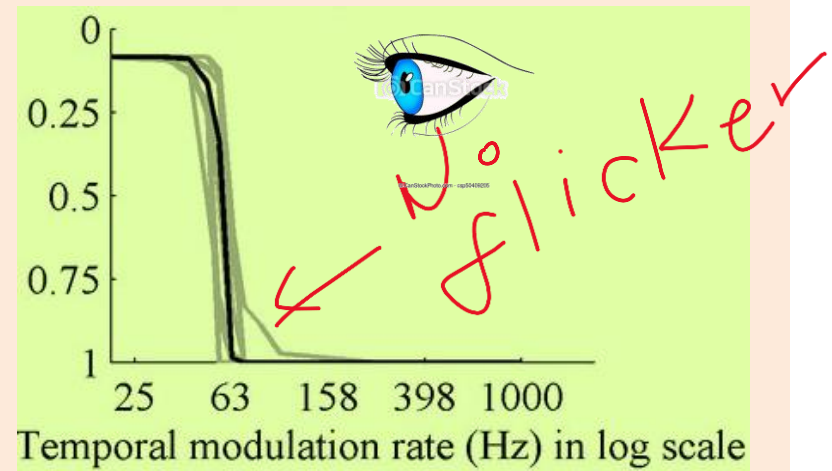
- Increases Eddy-current losses

$$P_e \propto B^2 f^2$$

- Increases cable and line impedance (inductive losses)

$$X = \omega L$$

- Too high frequencies become audible



Voltage levels and shape

- Common voltages:
 - 230V 50Hz - Norway and Europe
 - 110V 60 Hz - Canada, USA, Japan
 - Transmission (Norway)
 - 3.3, 6.6 11, 22, 33, 66, 132, 300, 420 kV
 - UHVAC today reaches 1000 kV in Russia

Mains electricity by country refers to

https://en.wikipedia.org/wiki/Mains_electricity_by_country

Review of Basic Concepts

Effective Value (Root Means Square value)

- Effective Value (RMS value) for sinusoidal voltage:

$$v = V_m \cos(\omega t + \phi).$$

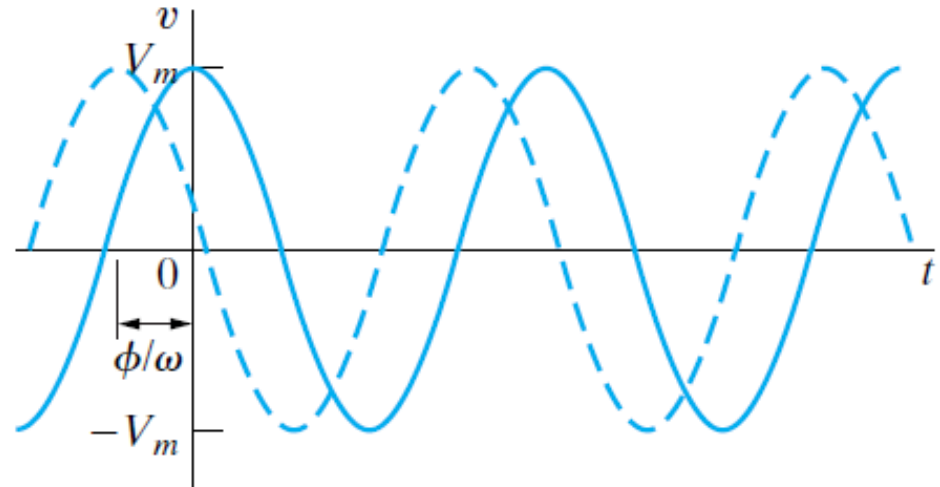
R

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt.}$$

M

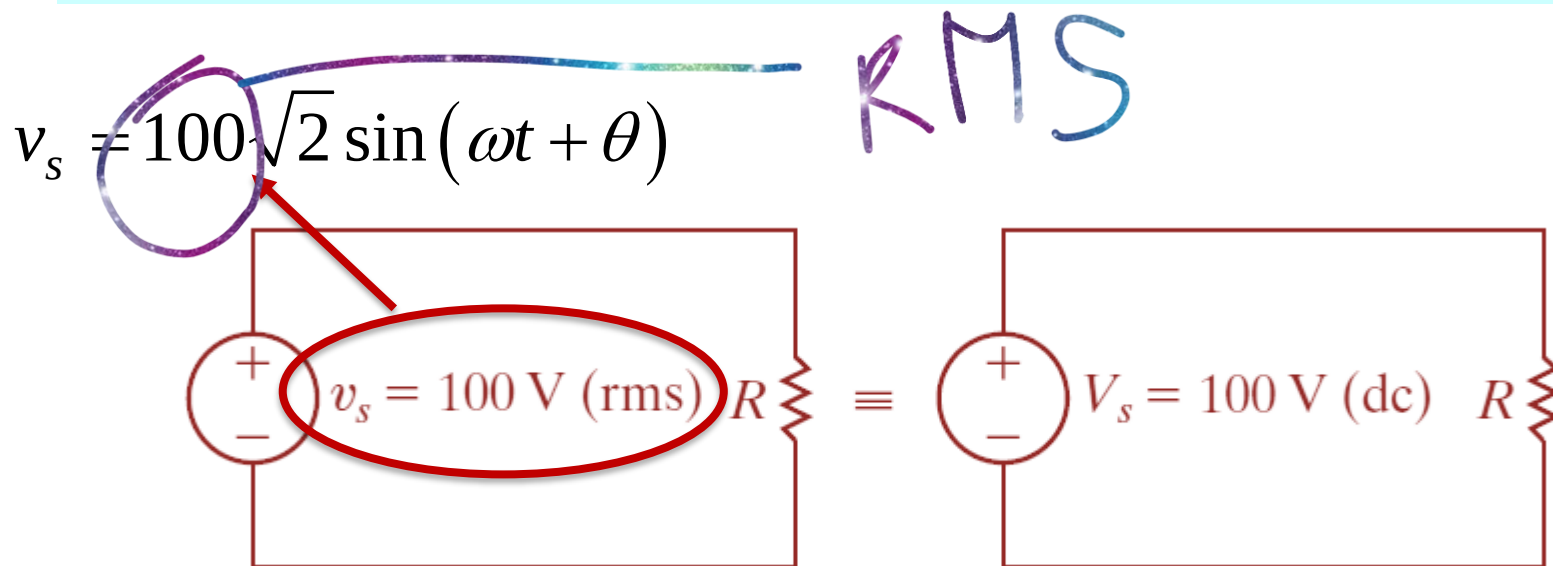
S

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}.$$



Physical interpretation of the RMS value

- AC system presented by RMS value is equivalent to a DC system in terms of average power delivered to the load.



Quick Review

Peak Value

A sinusoidal voltage is given by the expression:

$$v(t) = 325 \cos(100\pi t + \frac{\pi}{6}) \text{ V}$$

$$\omega = 2\pi f$$
$$f = 1/T$$

1. what is the frequency?

$$\frac{100\pi}{2\pi} = 50\text{Hz}$$

2. What is the period of the voltage?

$$\frac{1}{50} = 20\text{ms}$$

3. What is the value of the voltage at $t = 3.333\text{ms}$?

$$325 \cos\left(\frac{\pi}{2}\right) = 0$$

4. What is the effective value of the voltage?

$$\frac{325}{\sqrt{2}}$$

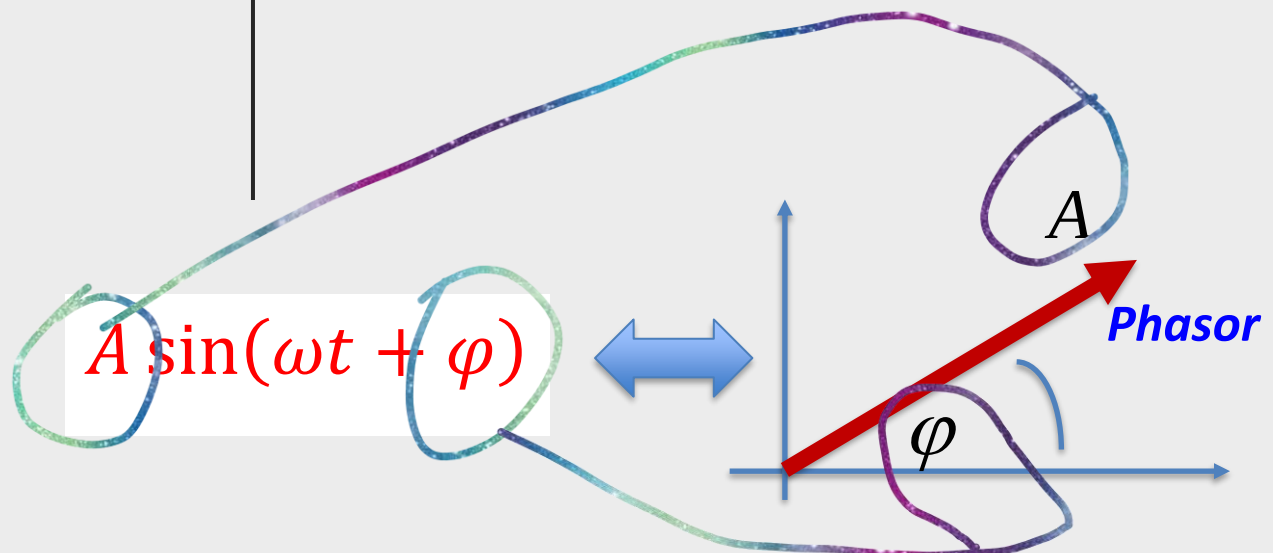
RMS

Phasor

Phasor

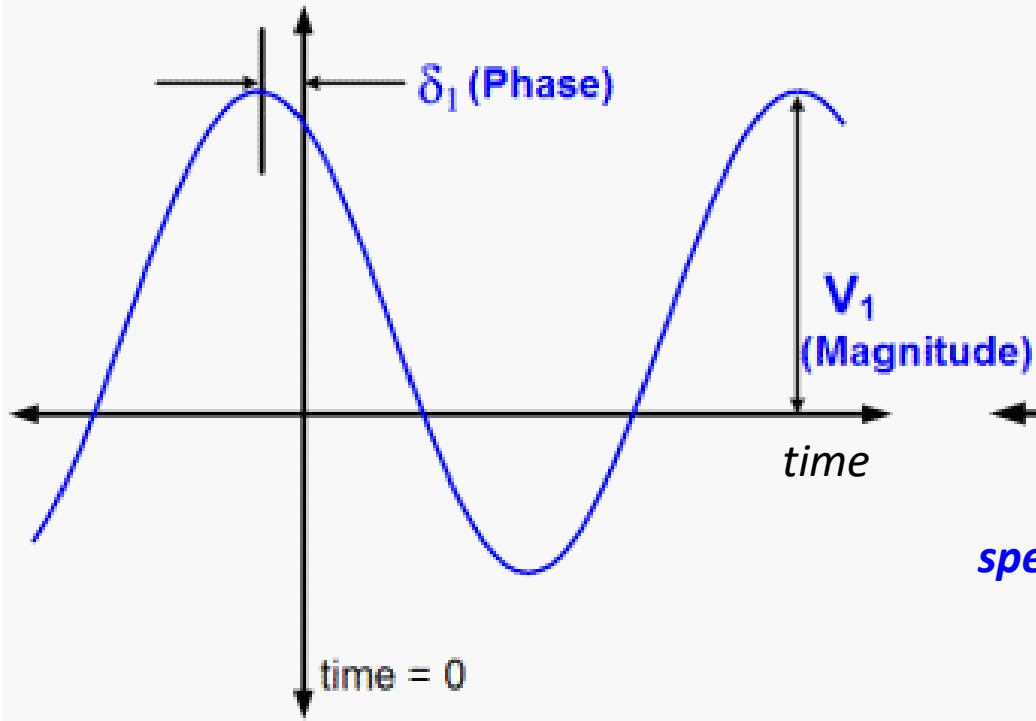
A complex number

- whose magnitude = magnitude of a corresponding sinusoid
- whose phase = phase of that same sinusoid.

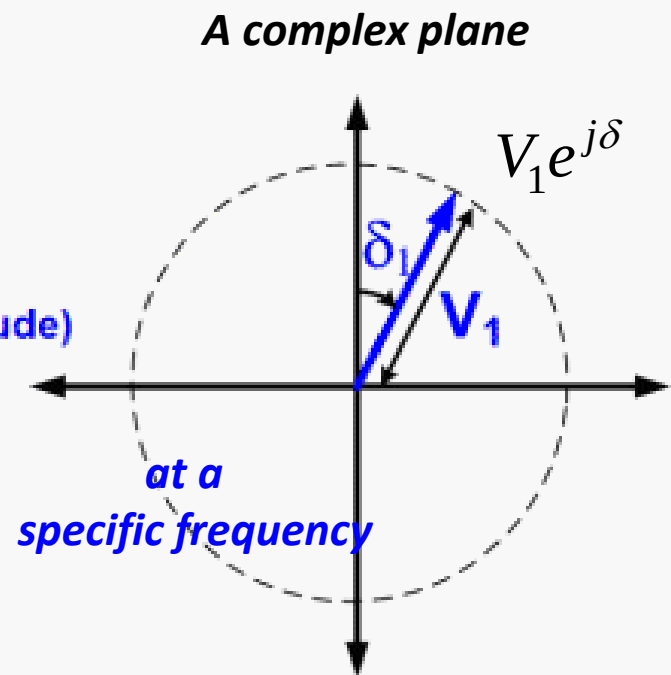


Sinusoid - Phasor

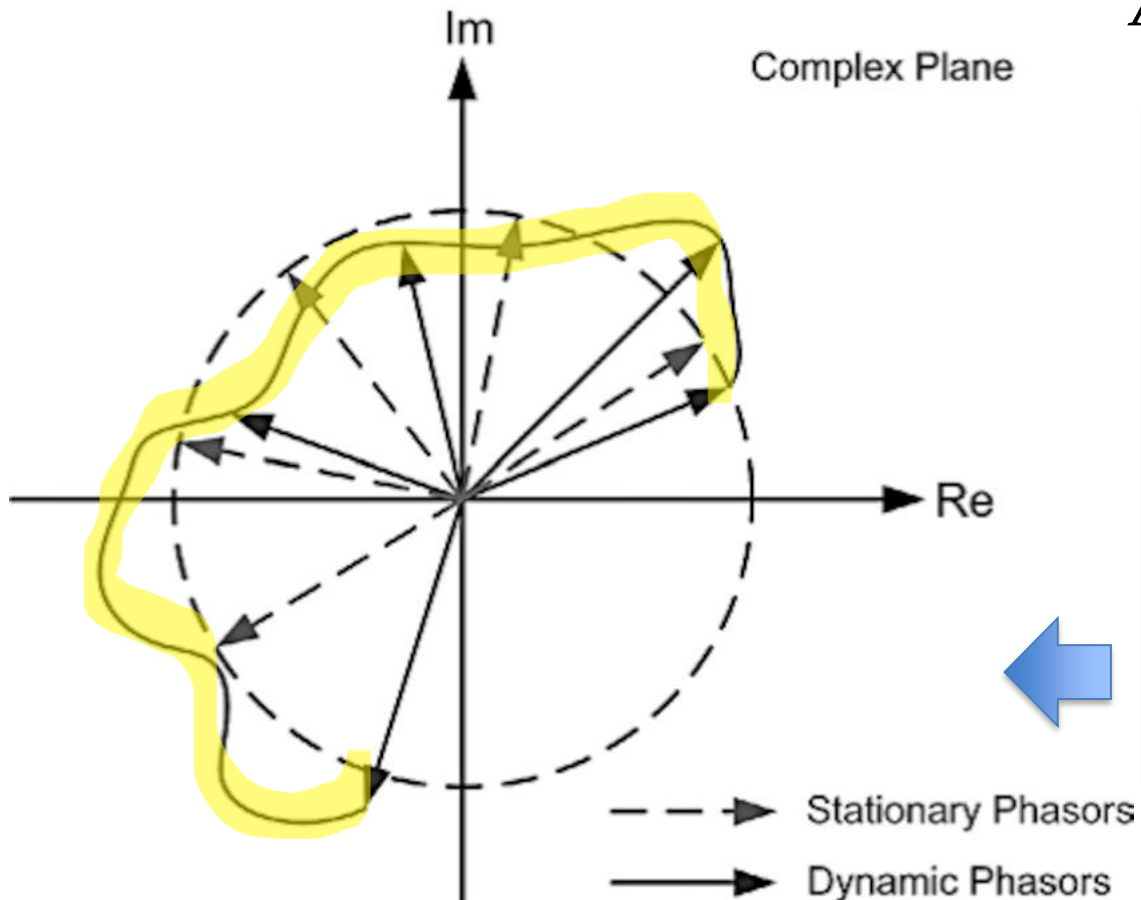
Sinusoidal Waveform



Phasor Representation



Dynamic - Phasor



$$A(t) \sin(\omega t + \varphi(t))$$

$A(t)$

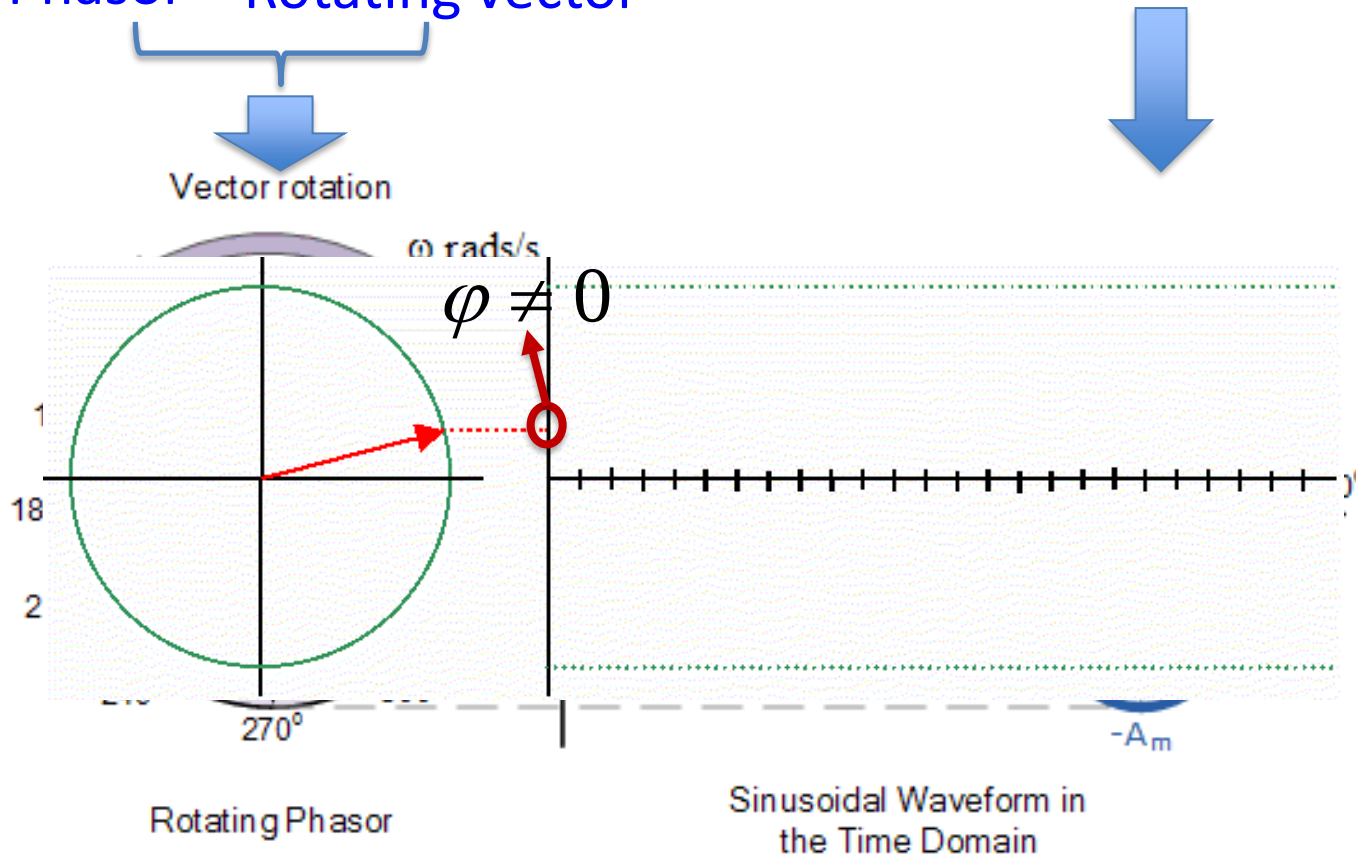
$\varphi(t)$

Time varying

*Time varying **Phasor***

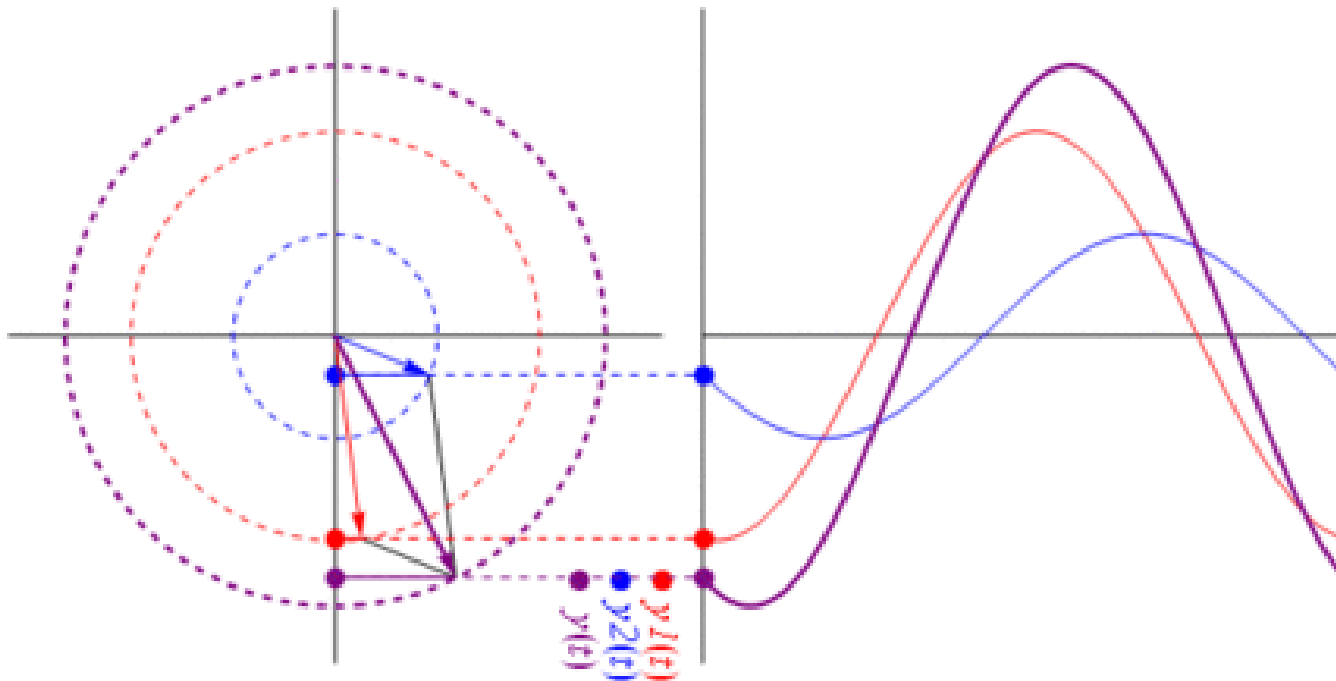
Phasor and sine wave

$$\underbrace{Ae^{j\varphi}}_{\text{Phasor}} \cdot \underbrace{e^{j\omega t}}_{\text{Rotating vector}} \xrightarrow{\text{Projects as}} A \sin(\omega t + \varphi)$$

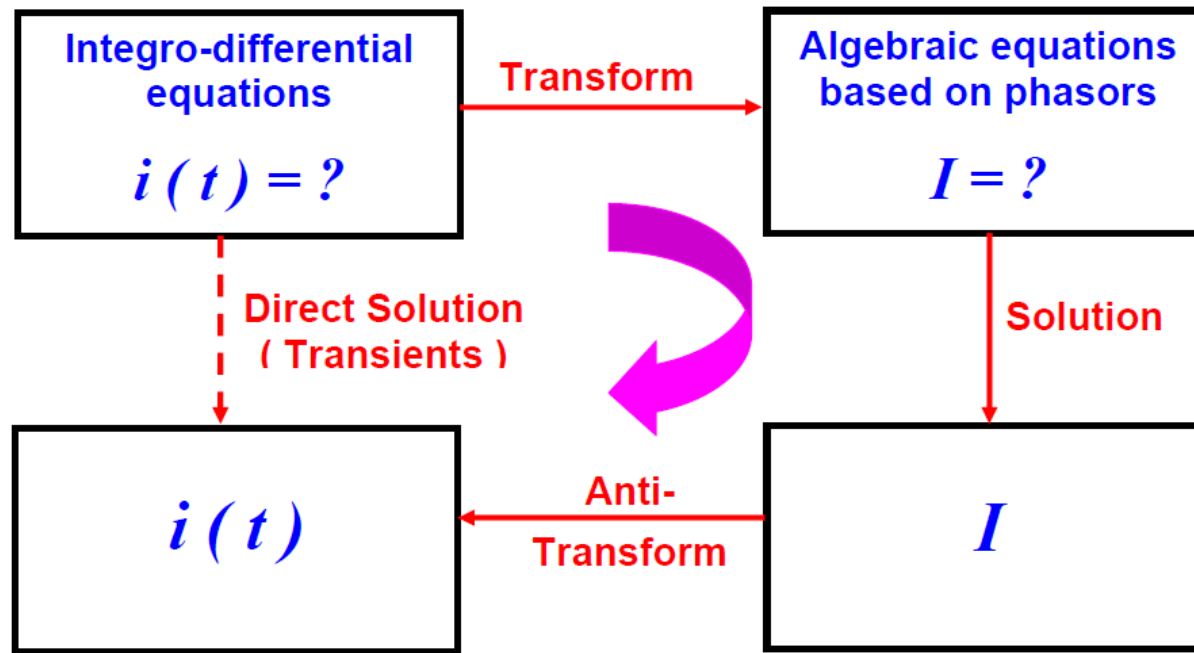


Phasors sum animation

$$\underbrace{\left(\underbrace{A_1 e^{j\delta_1}}_{\text{Phasor 1}} + \underbrace{A_2 e^{j\delta_2}}_{\text{Phasor 2}} \right)}_{\text{Phasor 1} + \text{Phasor 2}} \cdot \underbrace{e^{j\omega t}}_{\text{Rotating vector}} \xrightarrow{\text{Projects as}} A \sin(\omega t + \delta)$$

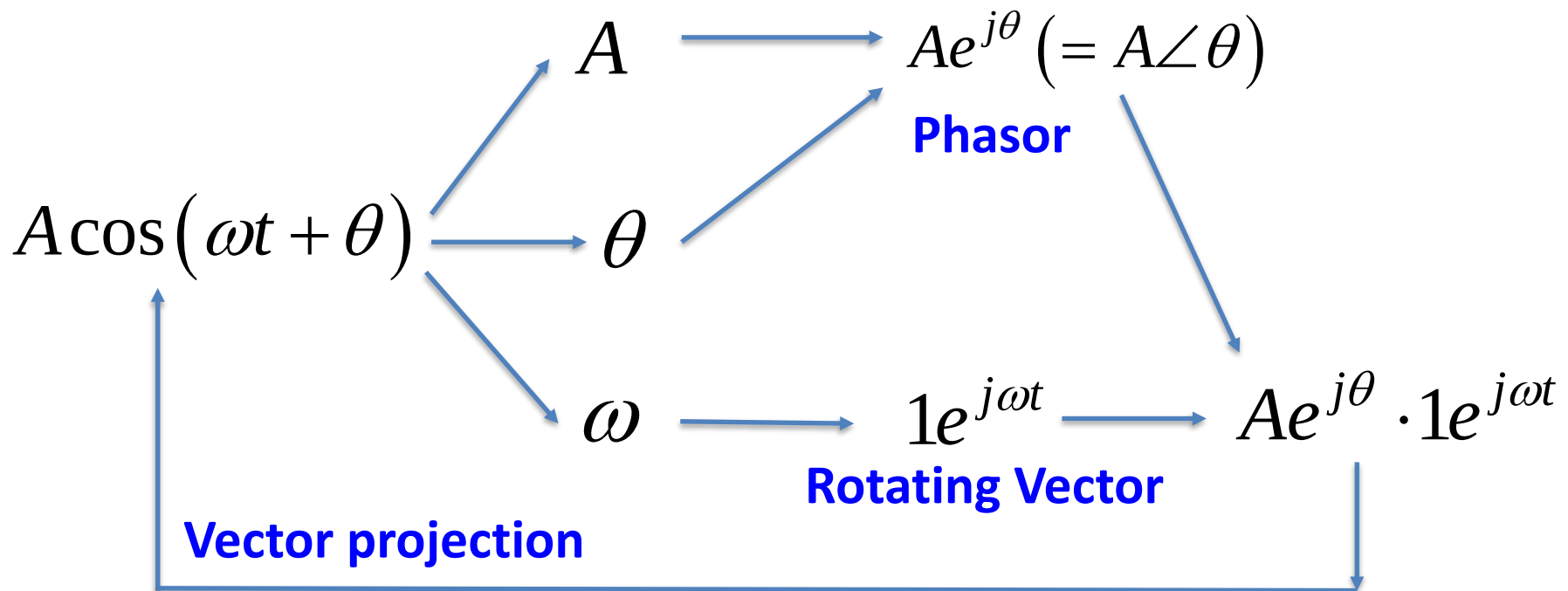


The phasor formalism provides a convenient way to solve **time-harmonic** problems in **steady state**, without having to solve directly a differential equation. The key to the success of phasors is that with the exponential representation one can immediately **separate frequency** and **phase** information. Direct solution of the time-dependent differential equation is only necessary for **transients**.



A summary of the phasor representation of a sinusoid

- A phasor is a complex number (can be time-varying) representing the amplitude and phase of a sinusoid.



What is the domain of phasors ?

- *Phasor* is in the frequency domain, because

$$Ae^{j\theta} \cdot 1e^{j\omega t}$$


Phasor is on top of a rotating vector, and the speed of the rotating vector is dependent on the frequency ω . From this we can find, the *phasor is actually defined at this specific frequency*, although this is not explicitly shown in the phasor formula.

$$\underline{Ae^{j\theta}} \text{ at } \omega$$

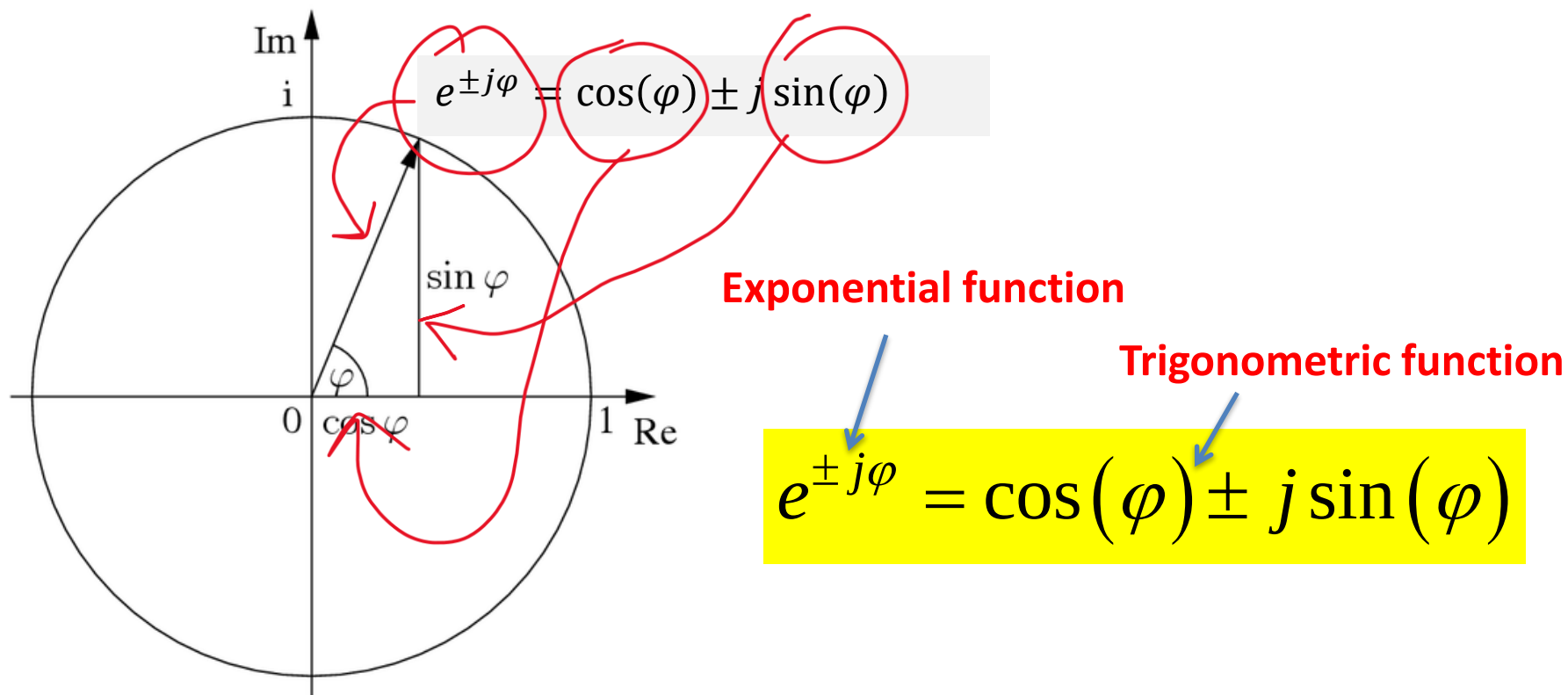
frequency domain



$$A \cos(\omega t + \theta)$$

Time domain

Review of the Euler identity

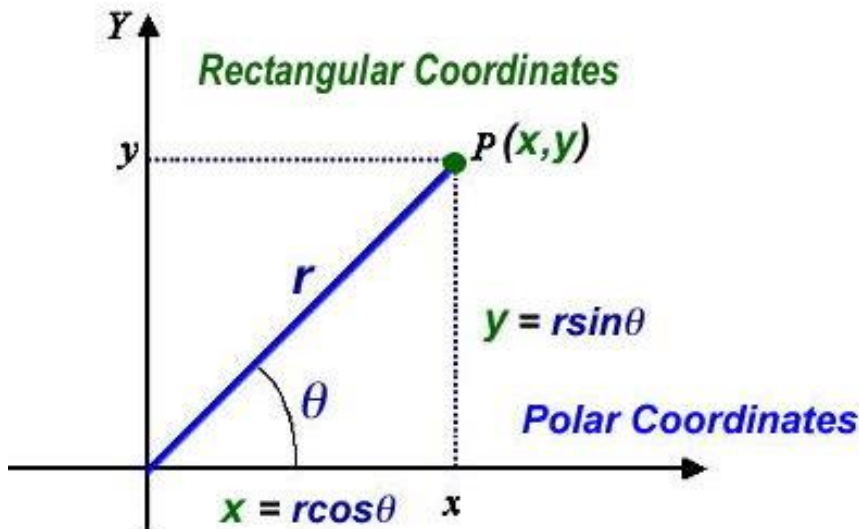


A complex-valued exponential can be represented by two real values in the complex plane

Coordinates for Phasor Diagram

$$re^{i\theta} = r \cos \theta + jr \sin \theta$$

Polar coordinate **Rectangular coordinate**



*Relationship between **Polar** and **Rectangular** coordinates*

$$x = \operatorname{Re} \{ re^{j\theta} \} = r \cos \theta$$
$$y = \operatorname{Im} \{ re^{j\theta} \} = r \sin \theta$$

Phasor-based analysis

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{-j\theta} = \cos\theta - j\sin\theta$$

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta$$

|||

$$e^{j\theta} + e^{-j\theta}$$

$$\cos\theta = \frac{\quad}{2}$$

Euler

Phasor-based analysis: *transform of a sinusoidal signal*

$$u = U_m \cos(\omega t + \theta)$$



Applying the *reformulated Euler identity*

$$u = \text{conj} \left\{ \frac{U_m e^{j\theta}}{2} \cdot e^{j\omega t} \right\} + \frac{U_m e^{j\theta}}{2} \cdot e^{j\omega t}$$



$$u = \frac{U_m e^{j\theta}}{2} \cdot e^{j\omega t}$$



Projection

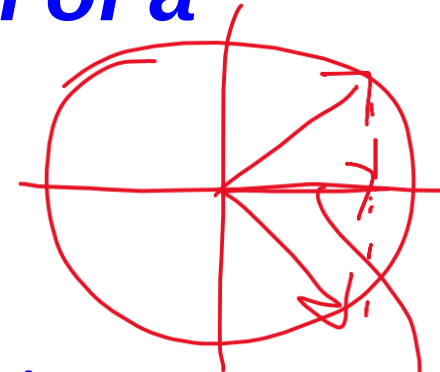
$$u = 2 \operatorname{Re} \left\{ \frac{U_m e^{j\theta}}{2} \cdot e^{j\omega t} \right\}$$



Original signal

$$u = U_m \cos(\omega t + \theta)$$

$$\frac{e^{j\theta} + e^{-j\theta}}{2} = \cos \theta$$



Scaled phasor

Rotating vector

Break

Phasor-based analysis: *calculation of the sinusoidal steady-state of a circuit*

Review: *A sinusoidal input to a linear system (circuit) will result in a magnitude-scaled and phase-shifted sinusoid at the output.*

$$A \cos(\omega t + \varphi_1) \Rightarrow H(j\omega) \Rightarrow A |H(j\omega)| \cos(\omega t + \varphi_1 + \angle H(j\omega))$$

Problem: *If we input a phasor into the linear system, what is the general format of the output?*

$$Ae^{j\theta} \text{ at } \omega \Rightarrow H(j\omega) \Rightarrow ?$$

Derivation I

$$U_m \cos(\omega t + \theta) \quad \Rightarrow \quad u = U_m e^{j\theta} \cdot e^{j\omega t}$$

Laplace

$$\Rightarrow U(s) = \frac{U_m e^{j\theta}}{s - j\omega} \quad \Rightarrow H(s) \Rightarrow Y(s) = H(s) \frac{U_m e^{j\theta}}{s - j\omega}$$

$$\Rightarrow Y(s) = \underbrace{\left(\frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \dots + \frac{K_m}{s - p_m} \right)}_{\text{Contributes to the transient response}} + \boxed{\frac{K_s}{s - j\omega}} \quad \text{Partial fractions of } Y(s)$$

The problem is reduced to find out K_s

Derivation II

$$Y(s) = \text{Partial fractions of } Y(s)$$

$$\left(\frac{K_1}{s-p_1} + \frac{K_2}{s-p_2} + \dots + \frac{K_m}{s-p_m} \right) + \frac{K_s}{s-j\omega}$$

Multiply both sides with $(s-j\omega)$ and do the limit

Contributes to the transient response

$$\begin{aligned} K_s &= \lim_{s \rightarrow j\omega} Y(s)(s-j\omega) \\ &= \lim_{s \rightarrow j\omega} \frac{H(s)U_m e^{j\theta}}{s-j\omega} (s-j\omega) \end{aligned}$$

$$K_s = H(j\omega)U_m e^{j\theta}$$

Derivation II

$$Y_{ss}(s) = \frac{K_s}{s - j\omega} \quad \Rightarrow \quad Y_{ss}(s) = \frac{H(j\omega)U_m e^{j\theta}}{s - j\omega}$$

Inverse Laplace

$$\Rightarrow y_{ss} = \boxed{H(j\omega)U_m e^{j\theta}} \cdot e^{j\omega t} \quad \text{Output phasor of the linear system}$$

$$\underbrace{U_m e^{j\theta}}_{\text{at } \omega} \quad \Rightarrow \quad \boxed{H(j\omega)} \quad \Rightarrow \quad \underbrace{H(j\omega)U_m e^{j\theta}}_{\text{at } \omega}$$

A summary

Input and output signal *

Input and output phasors

$$\left\{ \begin{array}{l} u = U_m e^{j\theta} \cdot e^{j\omega t} \\ y = H(j\omega) U_m e^{j\theta} \cdot e^{j\omega t} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} U = U_m e^{j\theta} \\ Y = H(j\omega) U_m e^{j\theta} \end{array} \right. \quad \text{at } \omega$$

$$\Rightarrow H(j\omega) = \frac{Y}{U} = \frac{\text{Output phasor}}{\text{Input phasor}} \quad \text{at } \omega$$

*Analytic signal since we only consider the positive frequency of a sinusoid

A clarification of transfer-functions

S-domain

$$H(s) = \frac{Y(s)}{U(s)}$$

$$s = j\omega$$



Frequency-domain

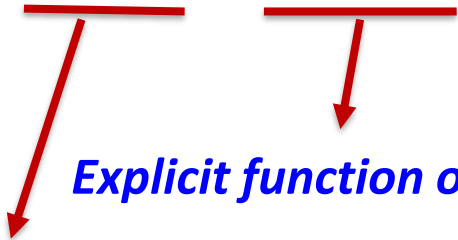
$$H(j\omega) = \frac{Y(j\omega)}{U(j\omega)}$$

We assume **input and output are sinusoids**

$$Y(j\omega) = Y \cdot e^{j\omega t}$$

$$U(j\omega) = U \cdot e^{j\omega t}$$

where the **frequency-domain input and output** can be explicitly represented by **phasors**.



Explicit function of w

Implicit function of w



$$H(j\omega) = \frac{Y(j\omega)}{U(j\omega)} = \frac{Y}{U}$$

Phasor-based transfer-function

Impedance/admittance definition using the phasors

$$\left. \begin{array}{l} V = V_m e^{j\varphi_v} \\ I = I_m e^{j\varphi_i} \end{array} \right\} \mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_m e^{j\varphi_v}}{I_m e^{j\varphi_i}} = \frac{V_m}{I_m} e^{j(\varphi_v - \varphi_i)} \quad \text{at } \omega$$

$$\left\{ \begin{array}{l} |\mathbf{Z}| = \frac{V_m}{I_m} \quad \text{Magnitude} \\ \angle \mathbf{Z} = \varphi_v - \varphi_i \quad \text{phase} \end{array} \right.$$

Admittance definition under phasor

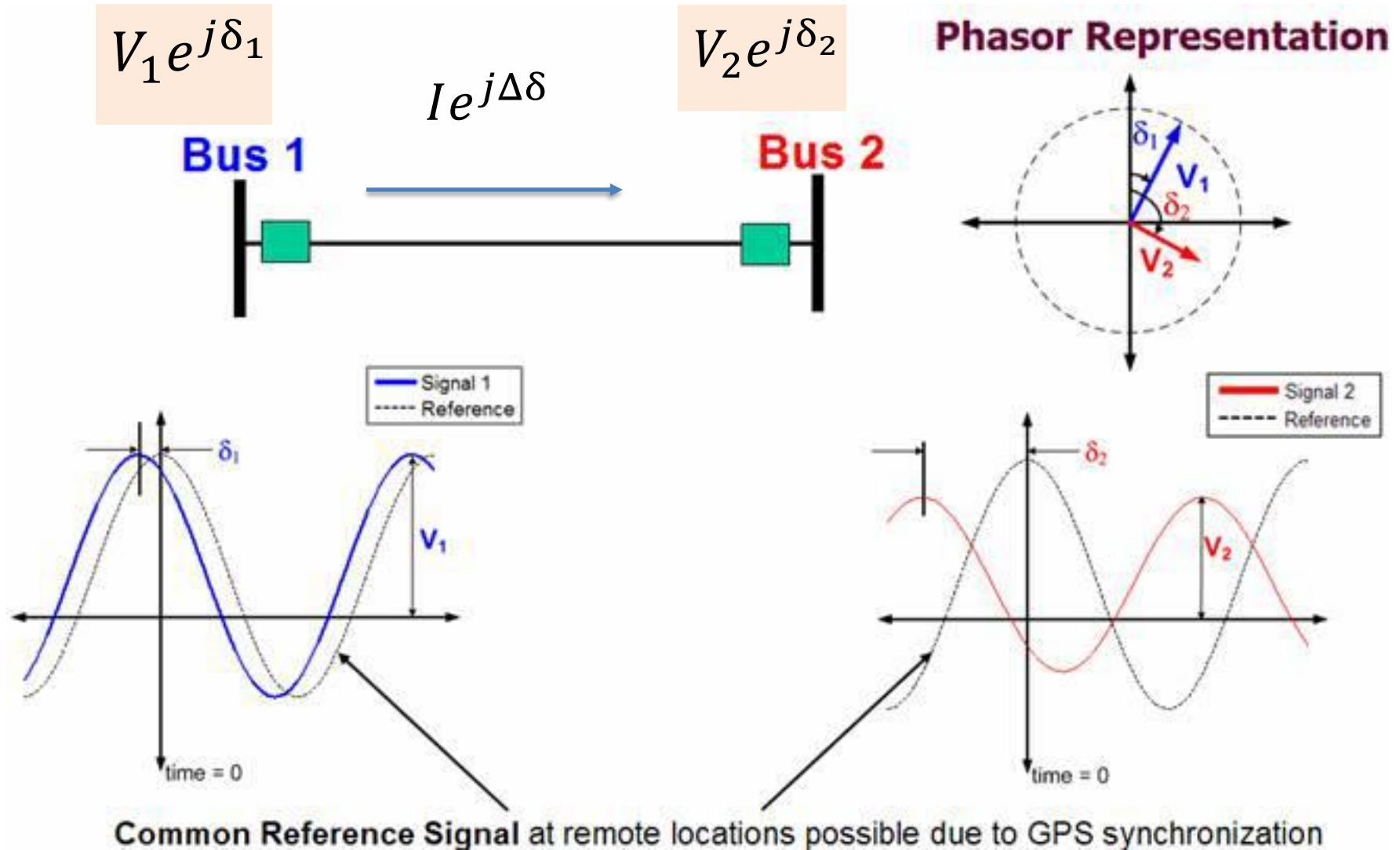
$$\mathbf{Y} = \frac{\mathbf{I}}{\mathbf{V}} = \frac{I_m}{V_m} e^{-j(\varphi_v - \varphi_i)}$$

Circuit elements

$$\mathbf{Z} = \frac{V_m}{I_m} e^{j(\varphi_v - \varphi_i)} = \frac{V_m}{I_m} \cos(\varphi_v - \varphi_i) + j \frac{V_m}{I_m} \sin(\varphi_v - \varphi_i)$$

$\varphi_v - \varphi_i = 0$	$\mathbf{Z} = \frac{V_m}{I_m}$	Resistor	$R = \frac{V_m}{I_m}$
$\varphi_v - \varphi_i = \frac{\pi}{2}$	$\mathbf{Z} = j \frac{V_m}{I_m}$	Inductor	$\omega L = \frac{V_m}{I_m}$
$\varphi_v - \varphi_i = -\frac{\pi}{2}$	$\mathbf{Z} = -j \frac{V_m}{I_m}$	Capacitor	$\frac{1}{\omega C} = \frac{V_m}{I_m}$

Where Phasors are used



Operations of complex number:

1. Addition

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

2. Subtraction

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

3. Multiplication

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$

4. Division

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$

1. Reciprocal

$$\frac{1}{z} = \frac{1}{r} \angle -\phi$$

2. Square root

$$\sqrt{z} = \sqrt{r} \angle \phi/2$$

3. Complex
conjugate

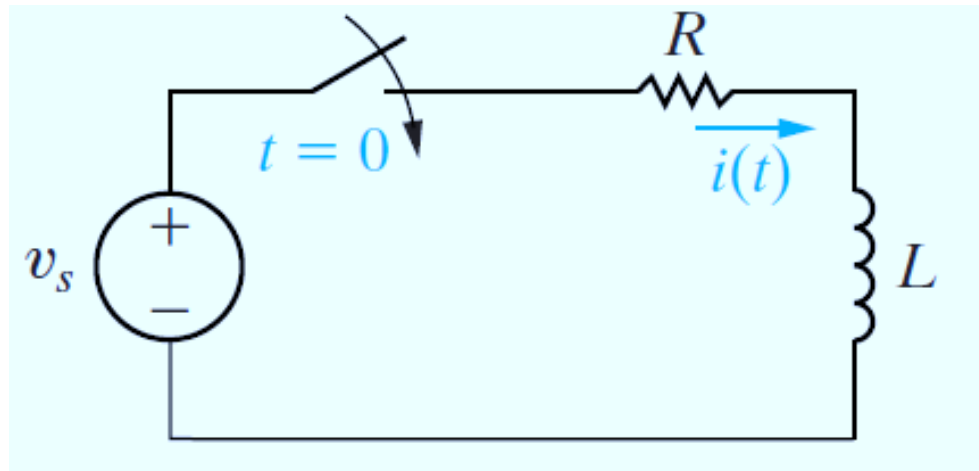
$$z^* = x - jy = r \angle -\phi = r e^{-j\phi}$$

4. Euler's identity

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

Example: phasor-based steady-state circuit analysis

$$v_s = V_m \cos(\omega t + \phi).$$

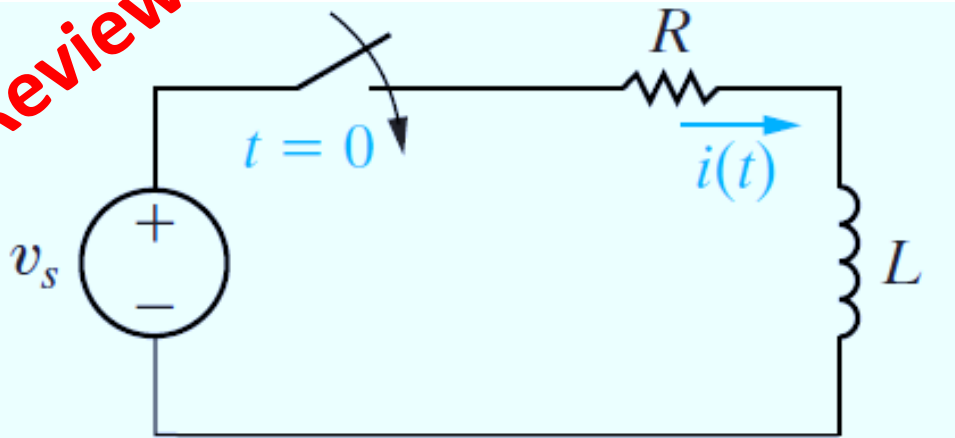


Problem:

Find the steady-state current response of the RL circuit?

Method 1: s-domain analysis

Review



$$v_s = V_m \cos(\omega t + \phi).$$

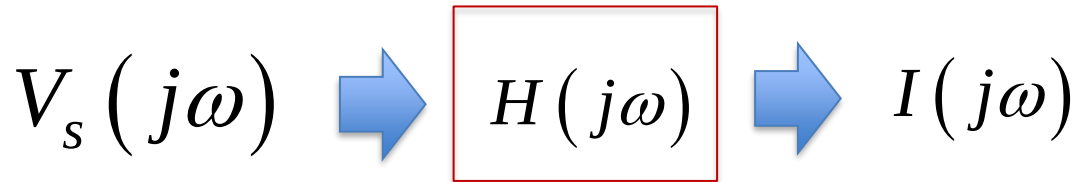
$$L \frac{di}{dt} + Ri = V_m \cos(\omega t + \phi),$$

Transient component

Steady State component

$$i = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta).$$

Method 2: Using the property of steady-state sinusoidal response



$$H(j\omega) = \frac{1}{R + j\omega L}$$

Magnitude

Phase

The diagram shows the steady-state response equation $i_{ss}(t) = V_m H(j\omega) \cos(\omega t + \varphi + \angle |H(j\omega)|)$. A red line originates from the $H(j\omega)$ term in the equation above and splits into two arrows. One arrow points to a red-bordered box around $H(j\omega)$ in the equation below, labeled "Magnitude". The other arrow points to a red-bordered box around $\angle |H(j\omega)|$ in the equation below, labeled "Phase".

$$i_{ss}(t) = V_m \boxed{H(j\omega)} \cos(\omega t + \varphi + \boxed{\angle |H(j\omega)|})$$

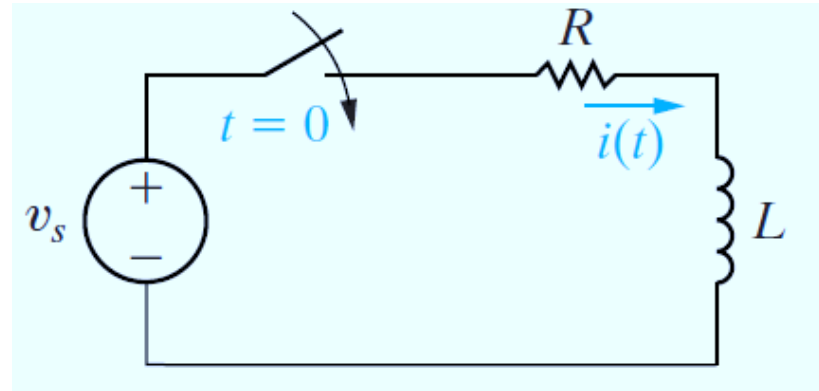
Method 3: Phasor-based analysis

$$v_s = V_m \cos(\omega t + \phi).$$

$$\mathbf{V}_s = V_m e^{j\phi}$$

$$H(s) = \frac{I(s)}{V_s(s)} = \frac{1}{R + sL}$$

$$\mathbf{I} = H(j\omega)\mathbf{V}_s = \frac{V_m e^{j\phi}}{R + j\omega L}$$



$$\mathbf{I} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} e^{j\left(\phi - \tan^{-1} \frac{\omega L}{R}\right)}$$

$$\text{Re}\left\{\mathbf{I} \cdot e^{j\omega t}\right\} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \phi - \tan^{-1} \frac{\omega L}{R}\right)$$

Properties of sinusoidal responses

Only when parameters R , L , and C are constant

- Steady-state solution = sinusoidal function
- Frequency of the response signal = frequency of the source signal.
- Phase angle of response signal $\neq$ phase angle of source

To find the steady-state response = to finding the maximum amplitude and phase angle of the response signal.

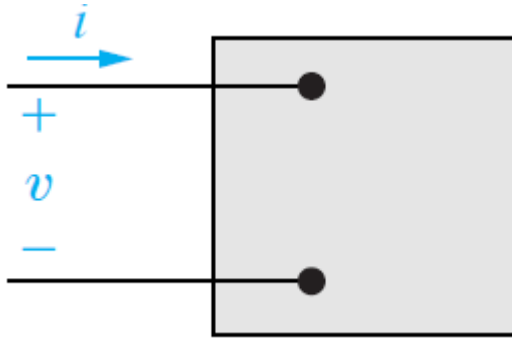
The waveform and frequency of the response are already known.

Sinusoidal Steady State Power

Sinusoidal Steady State Power

- Current, voltage and power calculations
 - Instantaneous power $p(t)$
 - Average (real) power P
 - Reactive power Q
 - Complex power S
 - Power factor pf

Instantaneous Power



$$v = V_m \cos(\omega t + \theta_v),$$

$$i = I_m \cos(\omega t + \theta_i),$$

Instantaneous power is defined as

$$p = vi.$$

$$p = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

Identity

$$\cos \alpha \cos \beta = \frac{1}{2} \cos (\alpha - \beta) + \frac{1}{2} \cos (\alpha + \beta)$$

$$p = V_m I_m \cos (\underbrace{\omega t + \theta_v}_{\alpha}) \cos (\underbrace{\omega t + \theta_i}_{\beta})$$



$$p = \frac{V_m I_m}{2} \cos (\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos (2\omega t + \theta_v + \theta_i)$$

Redefine the time at which the current phase crosses zero, i.e. $\omega t = \omega \tau - \theta_i$

$$p = \frac{V_m I_m}{2} \cos (\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos (2\omega \tau + \theta_v - \theta_i)$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Resubstitute
 $\tau = t$

$$p = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(2\omega t + \theta_v - \theta_i)$$

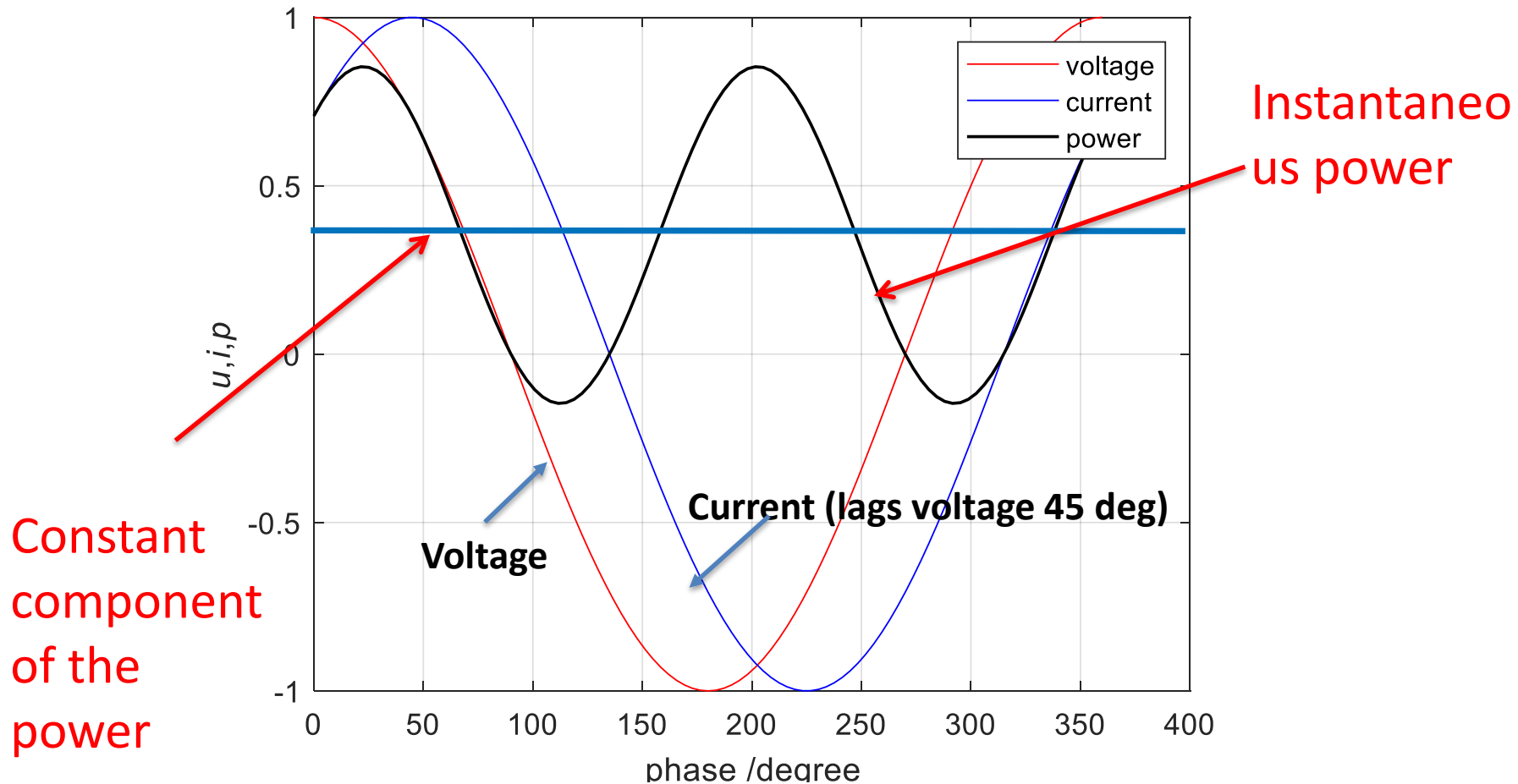


$$p = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \cos(2\omega t) - \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \sin(2\omega t)$$

constant

sinusoidal

Constant and oscillatory components



Average and Reactive Power

$$p = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \cos 2\omega t - \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \sin 2\omega t.$$

$$p = P + P \cos 2\omega t - Q \sin 2\omega t,$$


$$P = \frac{1}{T} \int_{t_0}^{t_0+T} p \, dt, \quad \text{Average value}$$

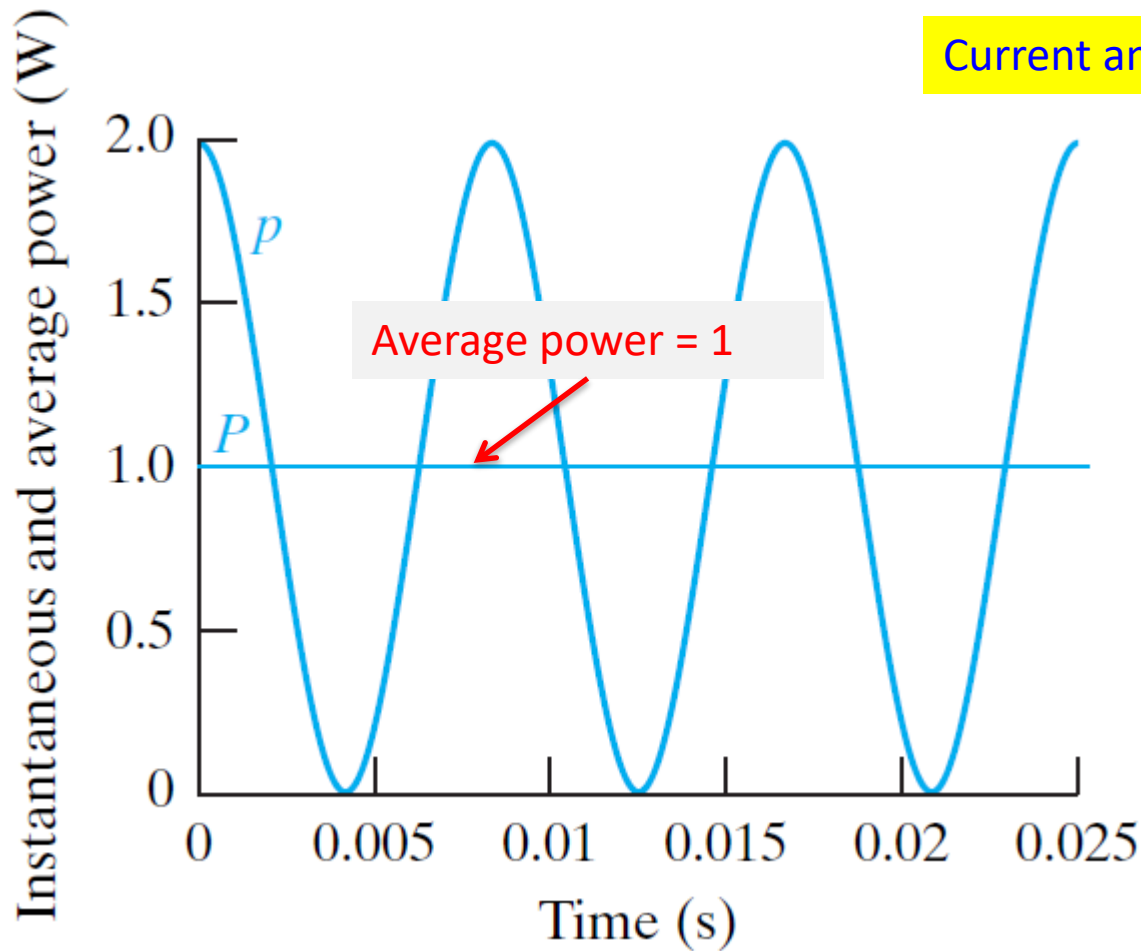
$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i),$$

Active power

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

Reactive power

Power for purely resistive circuits



Current and voltage are in phase

$$\theta_v = \theta_i.$$

$$p = P + P \cos 2\omega t.$$

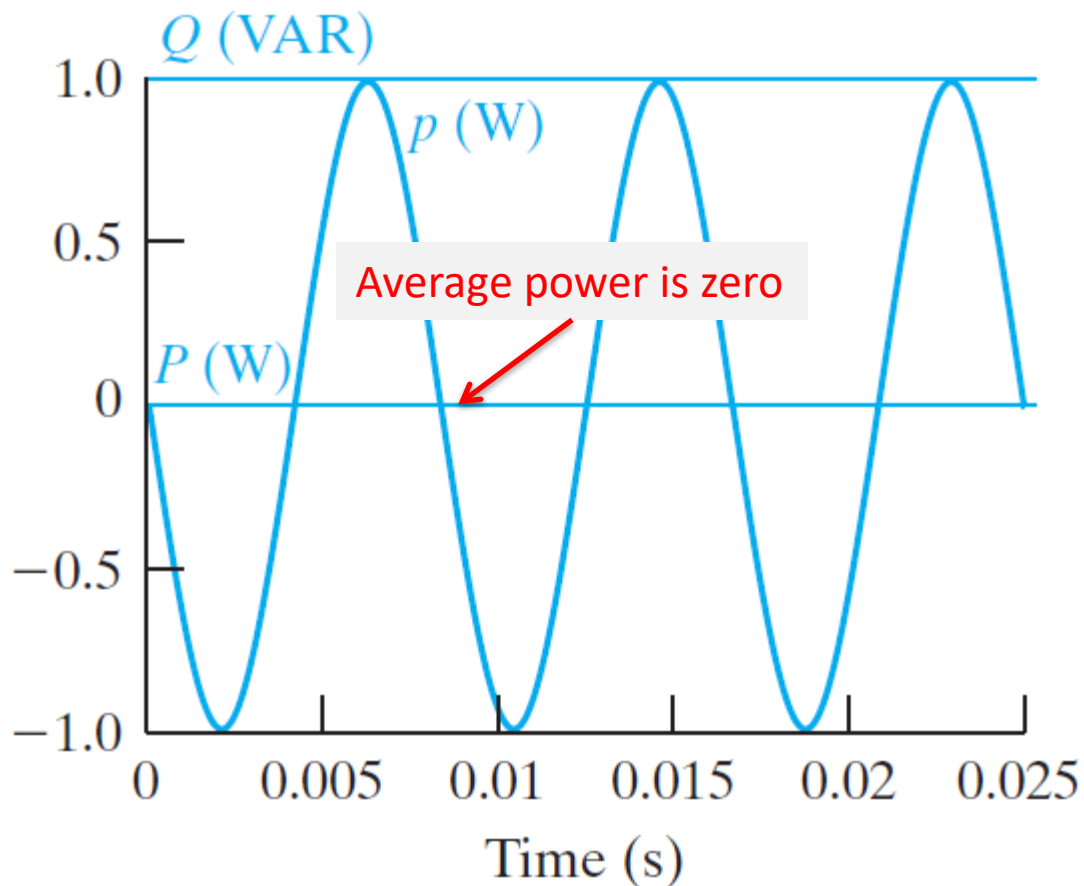
$$p = P + P \cos 2\omega t - Q \sin 2\omega t,$$

$$P = \frac{V_m I_m}{2} \cos (\theta_v - \theta_i),$$

Power for purely inductive circuits

Instantaneous, average and reactive power

Current lags voltage by 90°



$$\theta_v - \theta_i = +90^\circ$$

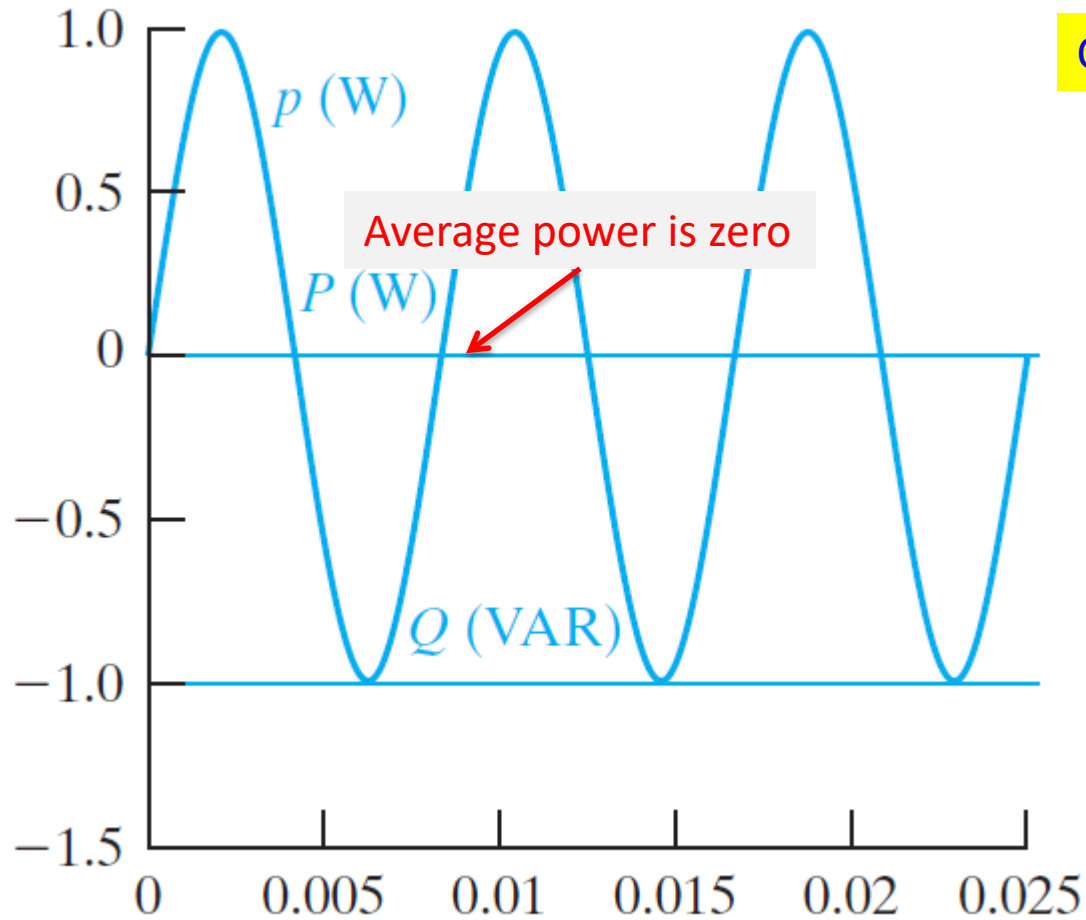
$$p = -Q \sin 2\omega t.$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

$$p = P + P \cos 2\omega t - Q \sin 2\omega t,$$

Power for purely capacitive circuits

Instantaneous, average, and reactive power



Current leads voltage by 90°

$$\theta_v - \theta_i = -90^\circ$$

$$p = -Q \sin 2\omega t.$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

$$p = P + P \cos 2\omega t - Q \sin 2\omega t,$$

Summary

- Note that the decision to use the current as the reference leads to Q being positive for inductors:

$$\theta_v - \theta_i = +90^\circ$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

Q is positive for L

- and negative for capacitors

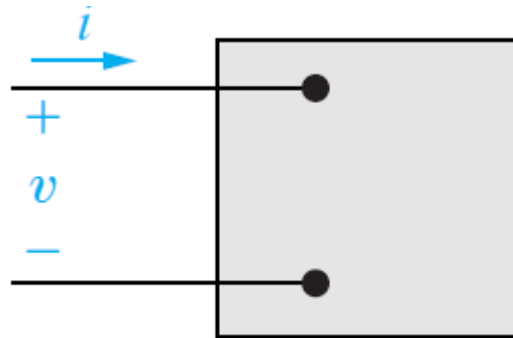
$$\theta_v - \theta_i = -90^\circ$$

$$Q = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i).$$

Q is negative for C

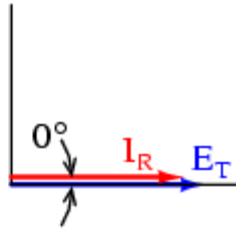
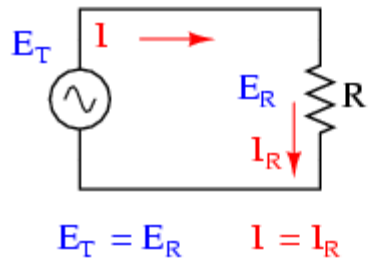
- Power engineers recognize this difference in the algebraic sign of Q by saying that ***inductors demand (or absorb) magnetizing vars, and capacitors furnish (or deliver) magnetizing vars.***

Delivered vs consumed p

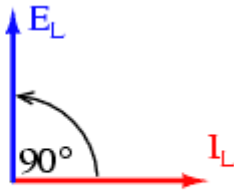
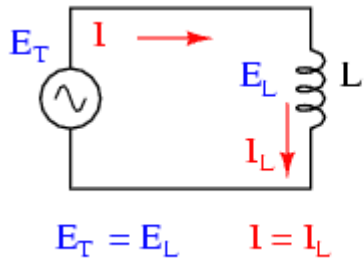
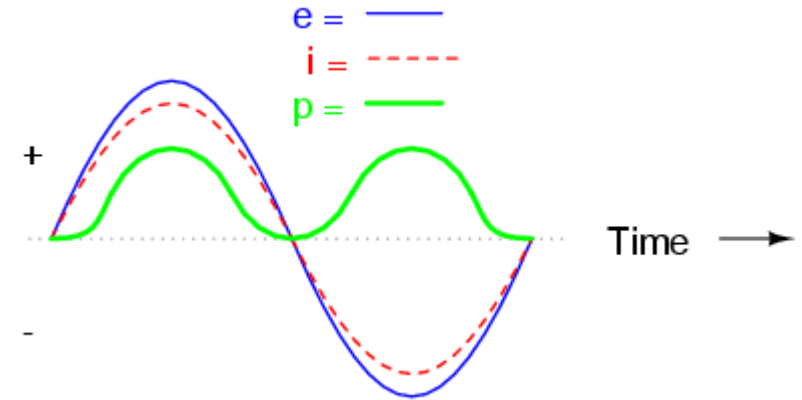


$$p = P + P \cos 2\omega t - Q \sin 2\omega t,$$

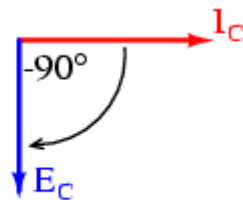
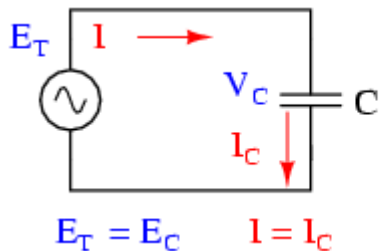
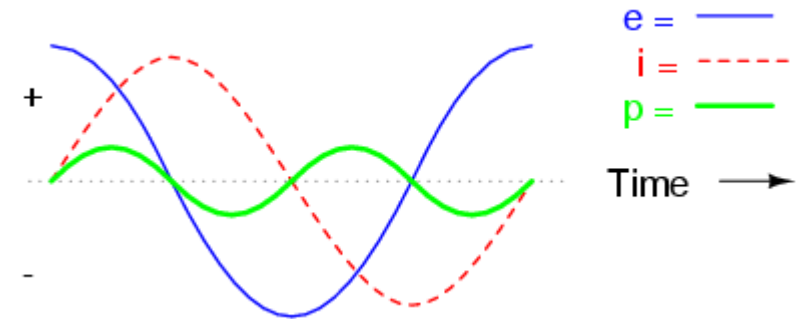
Power in the 3 passive elements



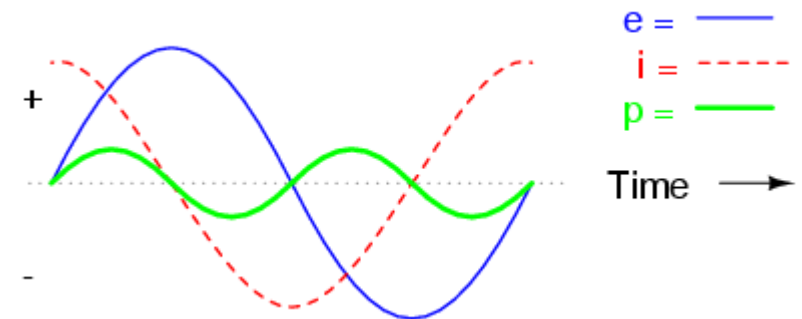
$$v_R = i \cdot R$$



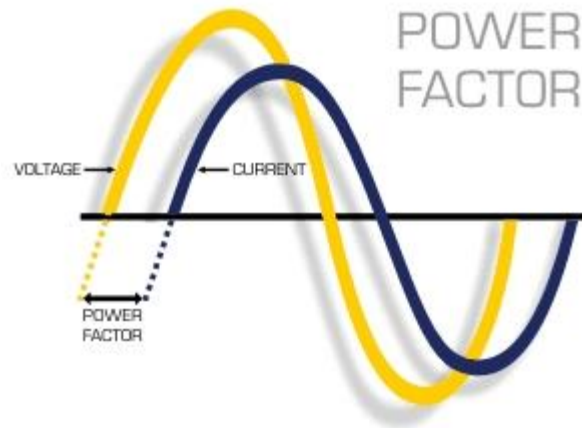
$$e_L = L \frac{di}{dt}$$



$$i_C = C \frac{de}{dt}$$



The Power Factor (PF)



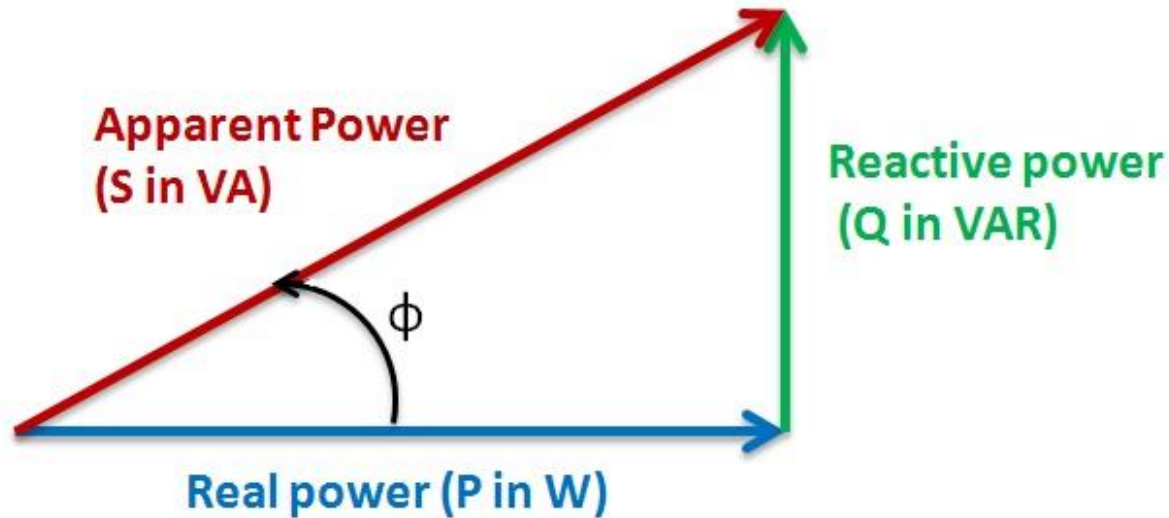
$$\text{pf} = \cos(\theta_v - \theta_i),$$

$$\left. \begin{aligned} P &= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \\ Q &= \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \end{aligned} \right\}$$

$$\sqrt{P^2 + Q^2} = \frac{V_m I_m}{2} = S$$

Apparent power

Power Factor

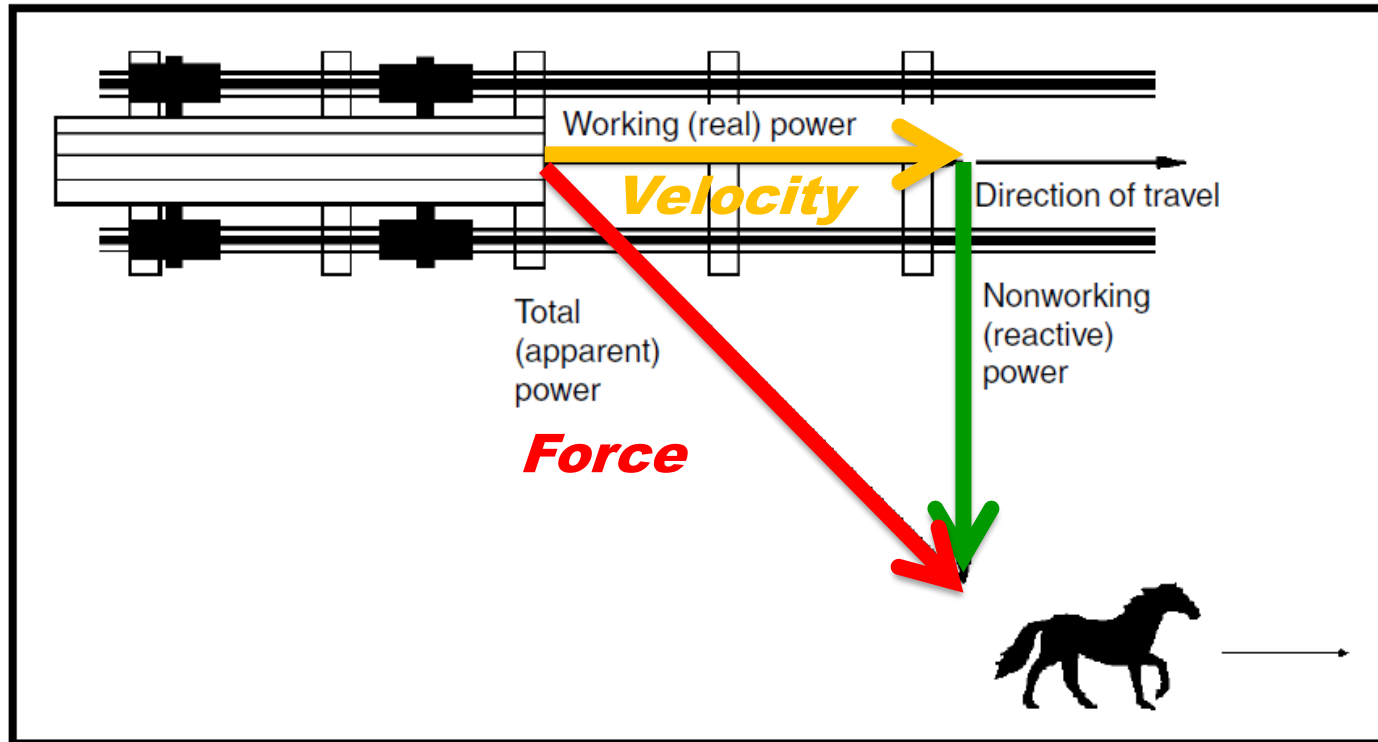


Power factor:

$$\cos \varphi = \frac{P}{S} = \frac{\frac{V_m I_m}{2} \cos(\theta_v - \theta_i)}{\frac{V_m I_m}{2}} = \cos(\theta_v - \theta_i)$$

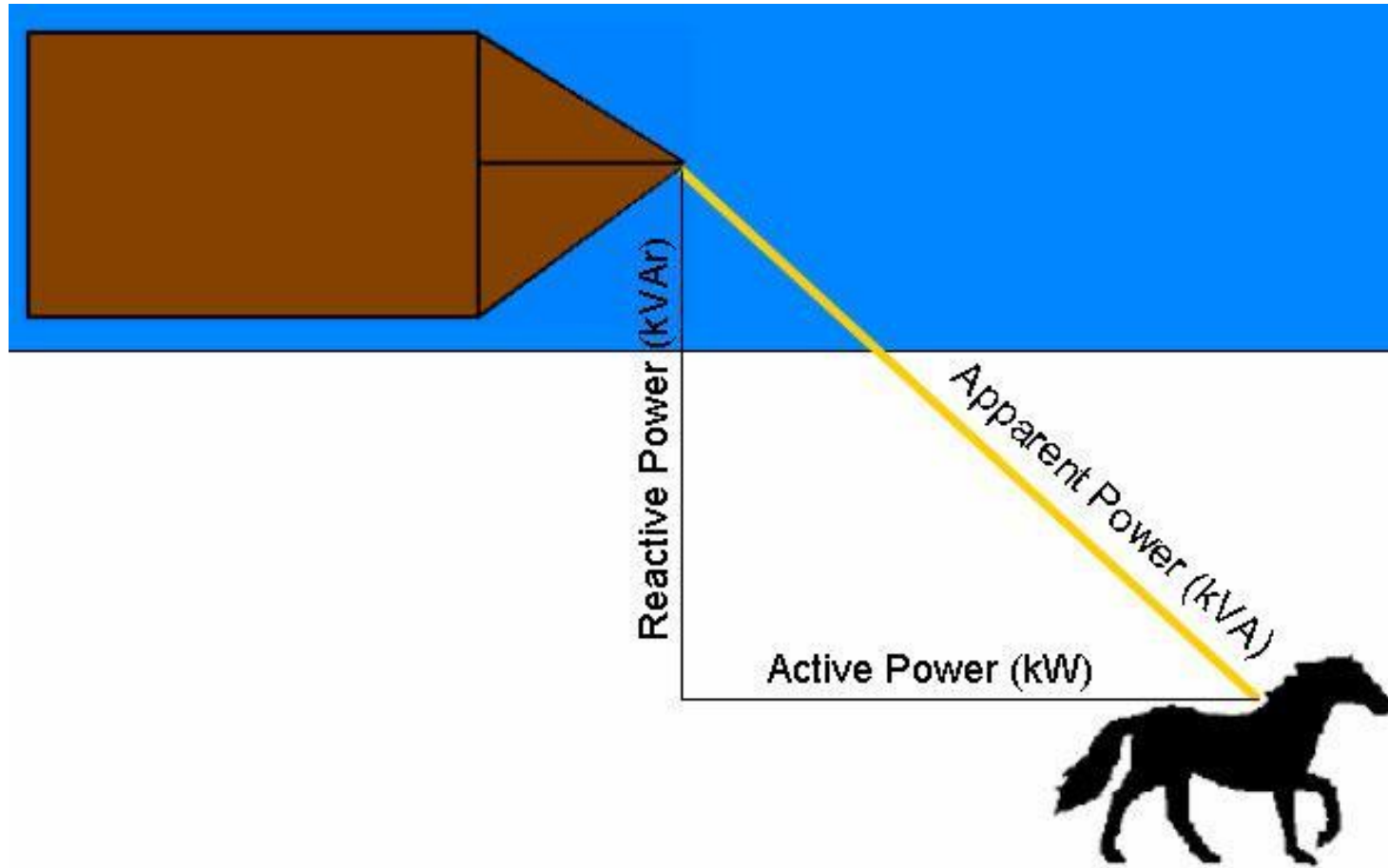
Analogies of Active/reactive Power

The Mechanical Analogy of Reactive Power

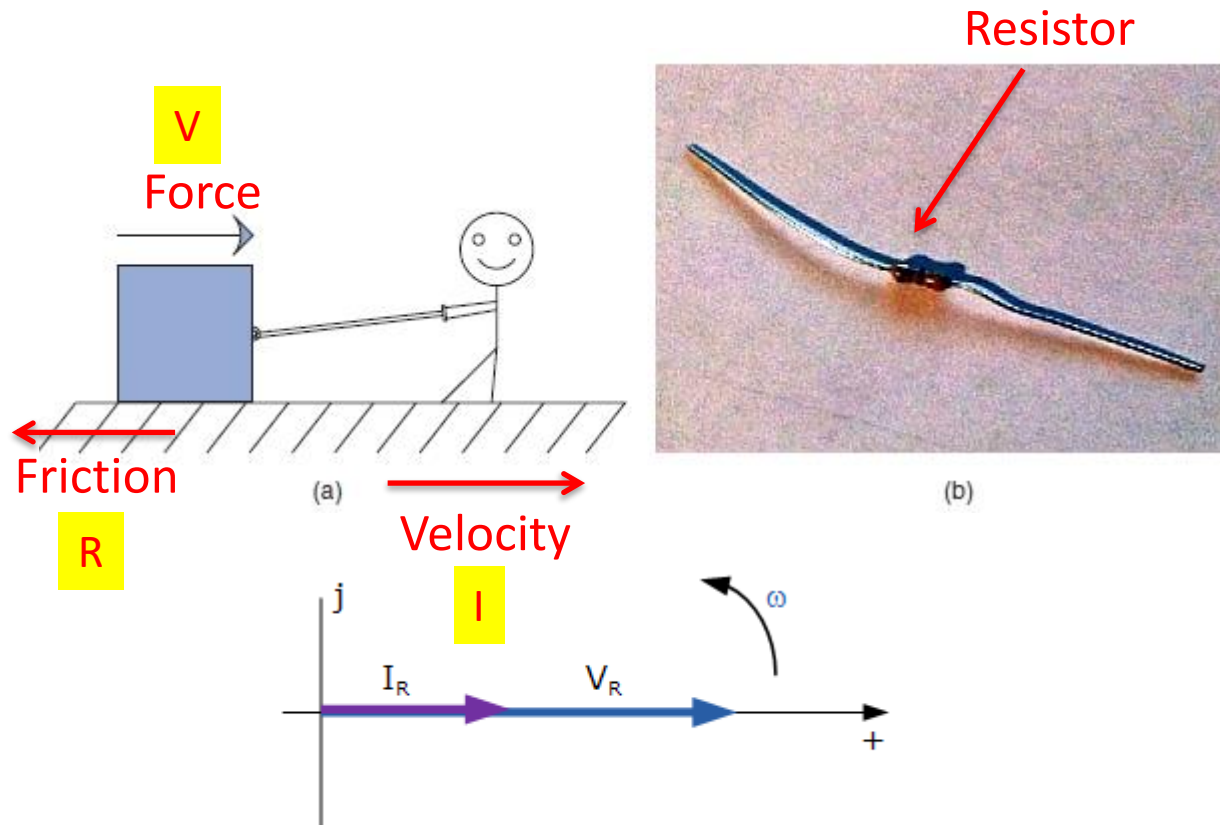


<http://lpsa.swarthmore.edu/Analog/ElectricalMechanicalAnalog.html>

Reactive Power Analogy

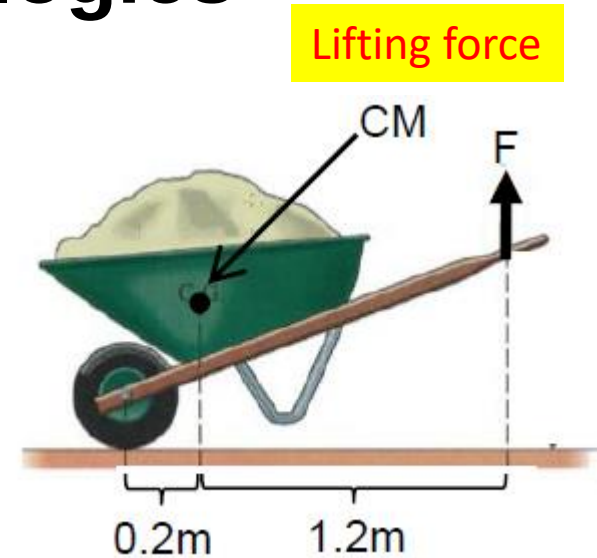
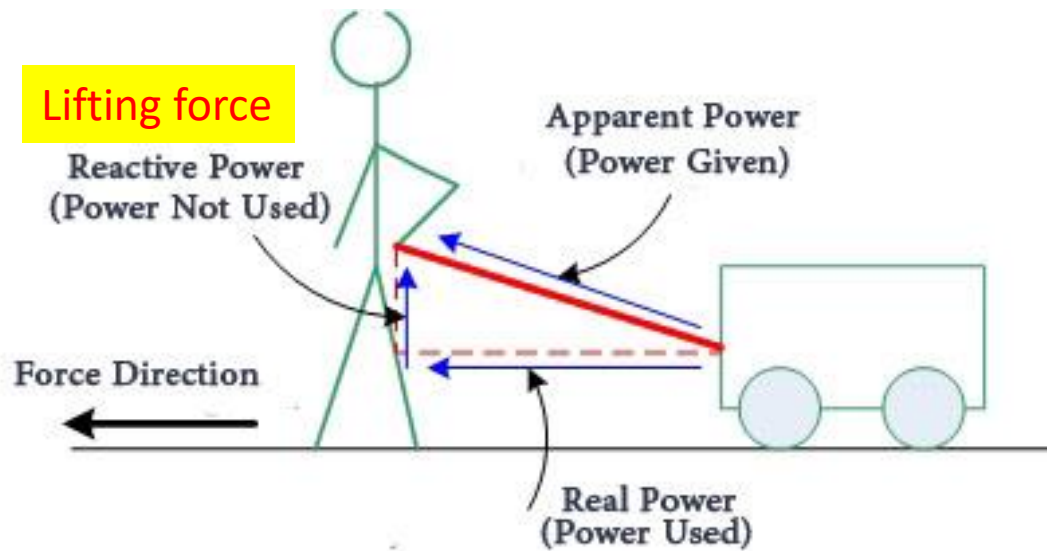


Analogy for resistive circuits

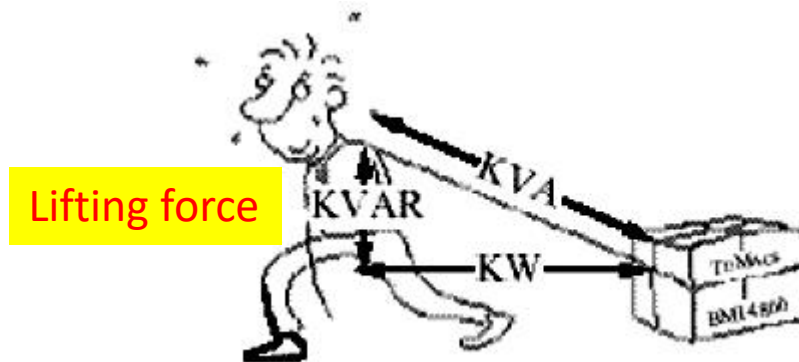
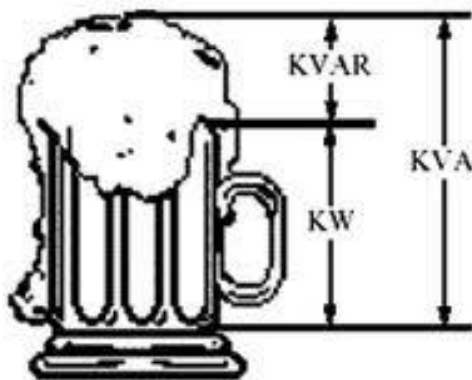


Resistance is considered as normal friction which will be in the same line with the force all the time.

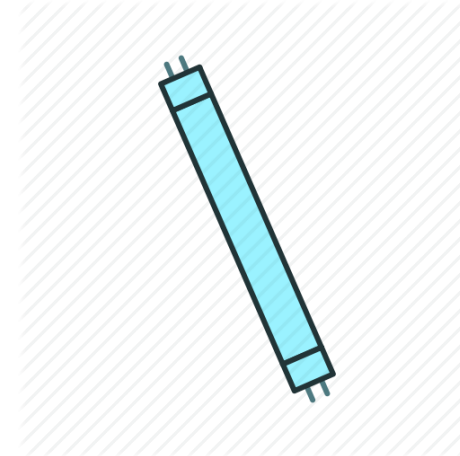
Reactive Power Analogies



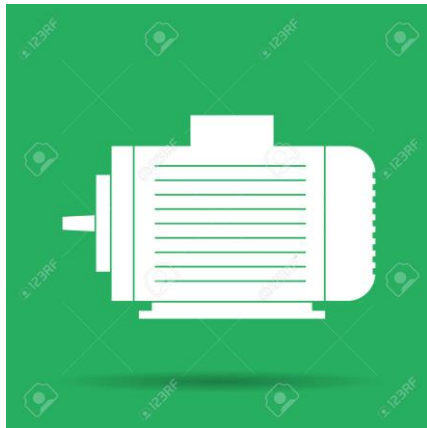
The Beer Analogy

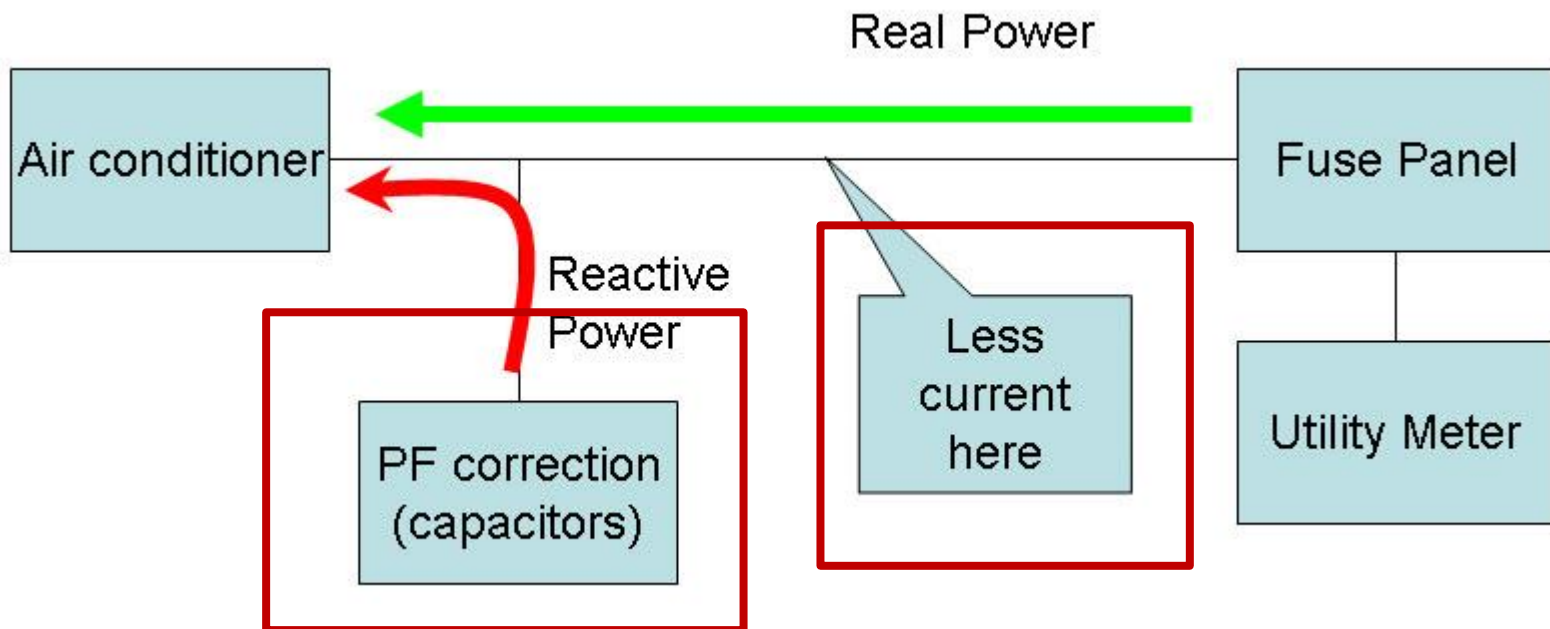
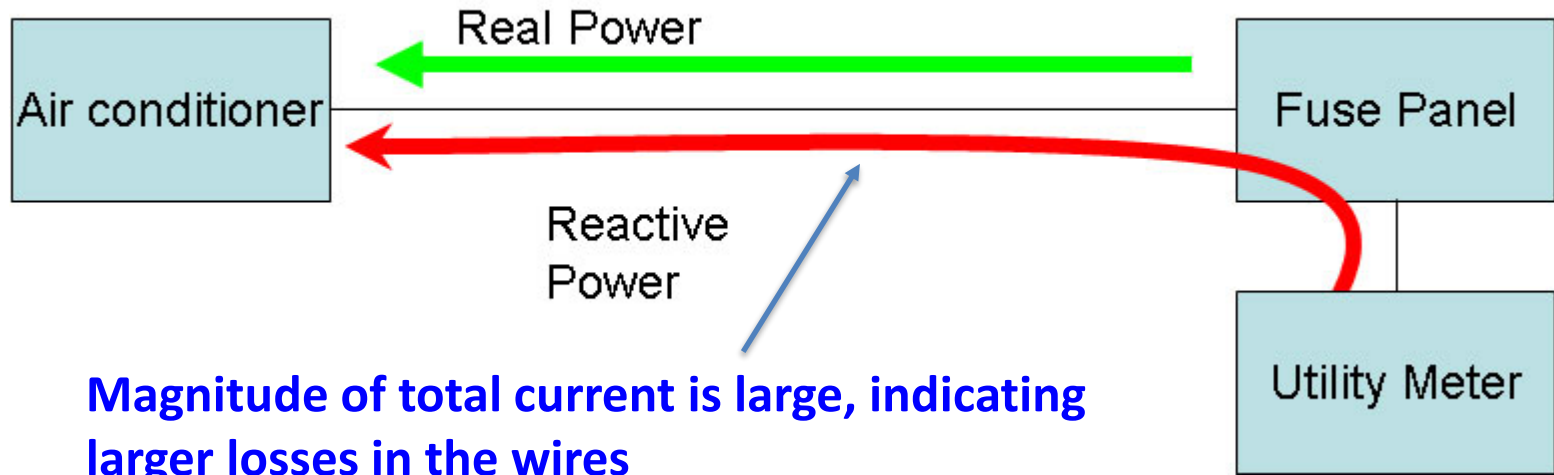


Lagging power factor loads

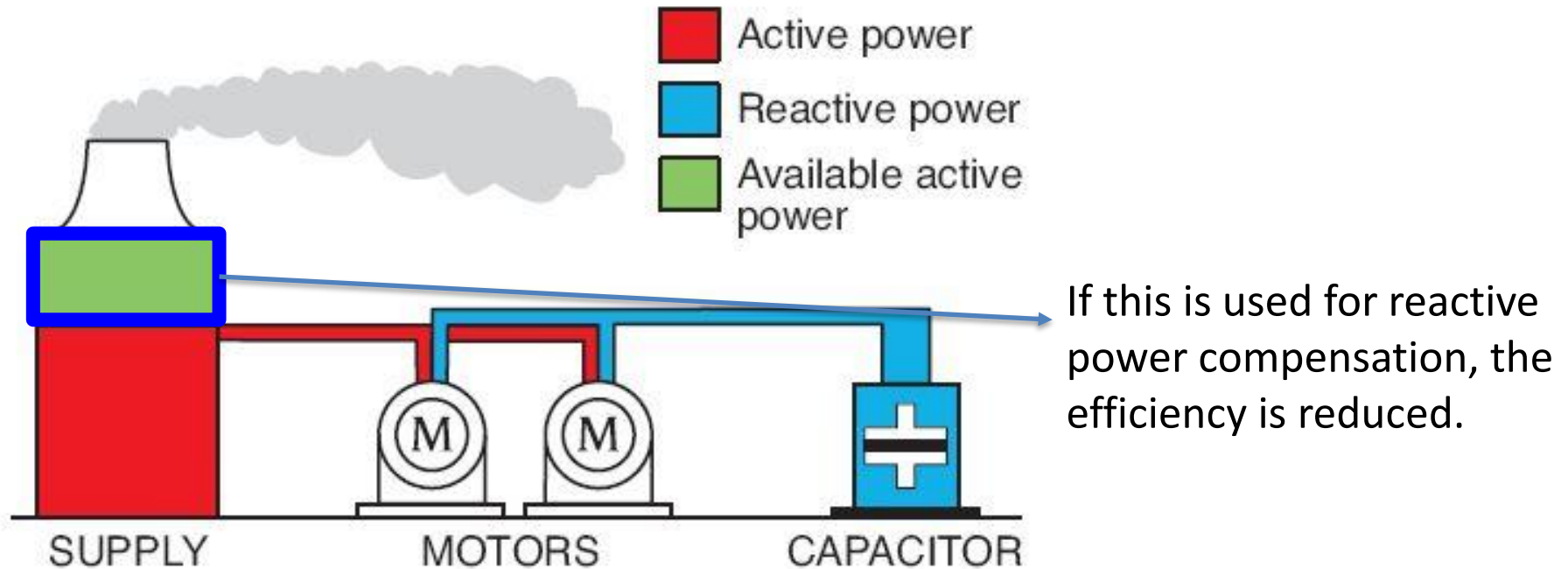


All of these have inductors or coils





Compensation of lagging power factor



The **efficiency** of the power plant is **maximized** if the reactive power of loads are compensated elsewhere.

End